NAJDA

NETWORK FOR AI-POWERED JOINT DISPATCH AND ASSISTANCE

Software Engineering Documentation

Requirements, User Stories, Use Cases & System Design

Version 2.0

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1 Introduction

1.1 Purpose of This Document

This document specifies NAJDA — the Network for AI-powered Joint Dispatch and Assistance — as delivered. It records the functional and non-functional requirements, user stories, use cases, architecture, data model, business rules, and verification approach for the platform, and traces each requirement to the component that satisfies it.

It is written for engineers who need to extend or operate the system, for reviewers who need to confirm what the platform does and why it is designed the way it is, and for anyone who needs a single authoritative description of the delivered behaviour. Every requirement in this document is implemented; the document describes the system that exists, not a plan.

1.2 Project Overview

NAJDA models how a citizen's emergency report becomes a coordinated, multi-party response. A citizen reports a Medical, Fire, or Police emergency with a location and optional evidence. The report is triaged with AI assistance, reviewed by a dispatcher, and assigned to a response unit that is genuinely on shift. The responding crew works the mission from a phone or a desk console, tracked live on a map with driving routes. For medical incidents, an ambulance crew selects a destination hospital and transmits live patient vitals to it. Every participant works from one shared incident record, scoped to what their role is permitted to see.

NAJDA is a simulation platform. It is not a real emergency service and is not connected to any government emergency system; submitting a report does not dispatch a real unit.

1.3 Objectives

- Model the full lifecycle of an emergency — report, triage, assignment, field response, hospital handoff, closure — on one traceable incident record.
- Keep humans in control: AI output (priority, duplicate suggestions) is advisory, dismissible, and never dispatches on its own.
- Make operational state trustworthy: a unit can only receive work if someone is on shift for it, and concurrent dispatch decisions cannot collide.
- Protect sensitive data: identity is delegated to a managed provider, free text is encrypted at rest, and media is only ever reachable through short-lived, authorization-checked links.
- Serve every actor in the form they actually work in: a web console for desk roles, a web and mobile experience for citizens and field responders, in English and Arabic.

1.4 Scope

In scope. Identity and access management; emergency reporting with evidence; dispatch operations; field response and shift management; hospital coordination with live vitals; AI-assisted triage; facility and unit administration; localization (English and Arabic, with right-to-left layout); a web console and a mobile companion application.

Operating context. The platform is scoped to Egypt: phone numbers use the +20 country code, incident coordinates are validated against Egypt's bounding box, and facility data is seeded from Egyptian OpenStreetMap records.

1.5 Development Methodology

NAJDA was built iteratively, module by module, in dependency order: identity and access first, then emergency reporting, dispatch, field response, and hospital coordination, followed by AI-assisted decision support, administration tooling, localization, and the mobile client. Each module's requirements, use cases, and design were recorded against the code that implements them, and this document is kept consistent with the delivered system.

1.6 Document Conventions

Term	Meaning
Priority	Must — essential to the workflow. Should — important, with a workable alternative. Could — convenience.
Status	Every requirement in this document is Implemented.
Platform	Where a use case differs between clients, it is marked Web (desk console and browser) or Mobile (native companion app).
LEAD / CREW	The two roles an employee can hold on a unit's shift. The LEAD is the unit's accountable operator.
Notation	UML use case, class, state, and sequence diagrams. `<<include>>` marks behaviour always performed; `<<extend>>` marks conditional behaviour; `<<triggers>>` (a practical extension) marks an automated system actor initiating a use case.

2 Stakeholders and System Actors

2.1 Human Actors

Actor	Responsibility	Client	Account creation
Citizen	Reports emergencies, adds evidence, chats with responders, follows the response, and may apply to become a First Responder.	Web, Mobile	Self-registration (passwordless email link or Google)
Dispatcher	Reviews the incident queue, assigns units, resolves duplicates, cancels assignments, and force-completes incidents.	Web	Provisioned by an Admin
Ambulance Crew	Works shifts and missions; for medical incidents selects a destination hospital and transmits vitals.	Web, Mobile	Provisioned by an Admin
Police · Firefighter	Works shifts and missions for police and fire incidents.	Web, Mobile	Provisioned by an Admin
First Responder	A vetted citizen who works missions from a personal unit.	Web, Mobile	Approved from a citizen's application, or provisioned
Hospital Staff	Monitors incoming patient transfers and live vitals for the linked hospital.	Web	Provisioned by an Admin, linked to a hospital
Admin	Manages accounts, facilities, units, incidents, and First Responder applications.	Web	Provisioned by an Admin
Super Admin	All Admin capabilities, plus protected operations: creating other Super Admins, overriding a phone number, deleting incidents, and reconciling Firebase accounts with database records.	Web	Provisioned by a Super Admin

Dispatcher, Hospital Staff, Admin, and Super Admin are desk roles served by the web console. If one of those accounts signs in to the mobile application it is shown the citizen experience rather than being blocked.

2.2 Login and Account Creation

Every account authenticates through Firebase Authentication; the NAJDA database remains the only source of truth for roles and permissions. Citizens register with a passwordless email link or Google. A citizen's database record is created only after their email address is confirmed, which prevents an unverified sign-up from reserving an address that belongs to someone else. Employee accounts are created by an administrator, who receives a one-time password-reset link to hand to the new employee. Section 6.1 describes both flows in detail.

2.3 System Actors

System actor	Role in the platform
AI Triage Service	Classifies incident priority and flags possible duplicate incidents, asynchronously, through a hosted language model.
Reverse Geocoder	Converts incident and facility coordinates into a readable address (OpenStreetMap Nominatim).
Identity Provider	Firebase Authentication — issues and verifies identity tokens, delivers sign-in and reset emails.
Object Storage	Supabase Storage — private bucket for evidence and application documents, reached only through signed URLs.
Routing and Map Services	OSRM driving routes, Overpass facility data, and OpenFreeMap tiles.

3 Functional Requirements

Requirements are grouped by module. Each row is implemented in the delivered system; the Requirements Traceability Matrix (Section 10) maps every requirement to its user story, use case, and implementing component.

3.1 Identity & Access Management

ID	Requirement	Actor(s)	Priority	Status
FR-1	A citizen shall be able to register with a passwordless email link or with Google. No account record is created until the email address is confirmed.	Citizen	Must	Implemented
FR-2	A citizen shall complete a profile (first and last name, phone, address, gender) and set a password before the account is considered complete; an incomplete profile gates use of the application.	Citizen	Must	Implemented
FR-3	Re-requesting a sign-in link for a partially registered account shall resume registration rather than fail.	Citizen	Should	Implemented
FR-4	The registration form shall indicate, as the citizen types, whether an email address already has an account.	Citizen	Could	Implemented
FR-5	Every API request shall be authenticated by a Firebase ID token verified on the server and resolved to a NAJDA user and role.	All	Must	Implemented
FR-6	An administrator shall be able to create an account for any non-citizen role; the system returns a one-time password-reset link. Only a Super Admin may create another Super Admin.	Admin, Super Admin	Must	Implemented
FR-7	Roles that operate from a facility shall be linked to a facility of the matching type (Hospital Staff must have one).	Admin, Super Admin	Must	Implemented
FR-8	An administrator shall be able to search, filter (by role and enabled state), and paginate the user list.	Admin, Super Admin	Must	Implemented
FR-9	An administrator shall be able to disable and re-enable an account. A Super Admin account is protected from plain Admins, and the last enabled Super Admin cannot be disabled.	Admin, Super Admin	Must	Implemented
FR-10	An administrator shall be able to change a user's role; the role claim on the identity record is re-synchronized.	Admin, Super Admin	Must	Implemented
FR-11	An administrator shall be able to override a user's email; a Super Admin shall be able to override a phone number, and every phone override is audit-logged.	Admin, Super Admin	Should	Implemented
FR-12	An administrator shall be able to generate a password-reset link for a user who signs in with a password.	Admin, Super Admin	Should	Implemented
FR-13	A phone number shall be unique among verified numbers only; unverified numbers may coexist, and a verified number can never be claimed by a second account.	System	Must	Implemented
FR-14	Users shall be able to edit their profile, change their email through a verification link, verify or change their phone by one-time code, and change their password after re-authenticating.	All	Should	Implemented
FR-15	Role-based access control shall be enforced on the server for every protected operation, independently of the client.	System	Must	Implemented

ID	Requirement	Actor(s)	Priority	Status
FR-16	A citizen with a verified email and phone shall be able to apply to become a First Responder with a motivation and at least one supporting document, submitted as a single request; only one application may be pending.	Citizen	Must	Implemented
FR-17	An administrator shall be able to review an application and its documents, approve it (role becomes First Responder and a personal unit is created) or reject it with a reason, and remove a document.	Admin, Super Admin	Must	Implemented
FR-18	A Super Admin shall be able to detect accounts that exist in Firebase but not the database or vice versa, create a profile from an orphaned Firebase account, and delete an orphan on either side.	Super Admin	Should	Implemented

3.2 Emergency Reporting

ID	Requirement	Actor(s)	Priority	Status
FR-19	A citizen shall be able to report an emergency by choosing a category (Medical, Fire, Police), optionally describing it and stating the number of injured people, and providing a location by device GPS or a manually placed pin within Egypt.	Citizen	Must	Implemented
FR-20	A citizen shall have at most one active incident at a time.	Citizen	Must	Implemented
FR-21	A citizen shall be able to attach photo, video, and audio evidence, uploaded directly to private storage through a signed URL and registered only after the file is confirmed to exist.	Citizen	Must	Implemented
FR-22	A citizen shall be able to edit the description and injured count and add or remove evidence until a unit is assigned; editing is then locked.	Citizen	Should	Implemented
FR-23	A citizen shall be able to view their active and past incidents; any user who participated in an incident may open its read-only history after it closes.	Citizen, Responder	Must	Implemented
FR-24	Evidence shall be viewable only through short-lived signed download URLs issued after an authorization check, with an in-page viewer.	All	Must	Implemented
FR-25	The system shall resolve a readable address for each incident from its coordinates, asynchronously and without delaying submission.	System	Should	Implemented
FR-26	Each incident shall have a live chat; only the citizen, dispatchers, admins, and responders on an active mission may send, while anyone who ever participated may read the history.	Citizen, Dispatcher, Responder	Must	Implemented
FR-27	Chat content, incident text, and patient-vitals notes shall be encrypted at rest.	System	Must	Implemented

3.3 Dispatch Operations

ID	Requirement	Actor(s)	Priority	Status
FR-28	A dispatcher shall see a live queue of incidents awaiting action and a live list of active incidents, updated without manual refresh.	Dispatcher	Must	Implemented
FR-29	A dispatcher shall be able to assign a unit by selecting it on the live map or in a list that can be filtered by unit type and is sorted nearest-first with an estimated arrival time, using the unit's live position or its home facility as a fallback.	Dispatcher	Must	Implemented
FR-30	Assignment shall be safe under concurrency: a unit can be offered to only one incident at a time, a unit becomes busy when offered, and several units may work the same incident.	System	Must	Implemented
FR-31	A dispatcher shall be able to cancel an assignment at any non-terminal stage, which frees the unit and reopens the incident if no other unit is working it.	Dispatcher	Must	Implemented
FR-32	A dispatcher shall be able to force-complete an incident, completing every active mission and freeing every unit.	Dispatcher	Must	Implemented
FR-33	A Super Admin shall be able to permanently delete an incident and its dependent records, freeing any unit still tied to it.	Super Admin	Should	Implemented
FR-34	The live operations map shall show the incident, every unit working it, and a colour-coded route per mission, switchable between driving and straight-line.	Dispatcher	Must	Implemented
FR-35	A dispatcher shall be able to mark an incident as a duplicate of another, and to confirm or dismiss an AI-suggested duplicate; the cancellation reason, actor, and time are recorded.	Dispatcher	Must	Implemented
FR-36	A dispatcher shall be able to retry AI priority classification for an incident that has none.	Dispatcher	Should	Implemented

3.4 Field Response (Mission Lifecycle)

ID	Requirement	Actor(s)	Priority	Status
FR-37	An employee shall be able to start a shift on a unit as LEAD, join a unit's shift as CREW, leave as CREW, or end the shift as LEAD; an employee may hold one active shift, and eligibility follows facility rules.	Responder	Must	Implemented
FR-38	A First Responder shall work only from their own personal unit.	First Responder	Must	Implemented
FR-39	Offering a mission shall snapshot the crew on shift; only the unit's LEAD may accept or reject it, and rejection frees the unit.	Responder	Must	Implemented
FR-40	The LEAD shall be able to mark a mission en route and arrived, complete a non-ambulance mission, and withdraw while the mission is accepted or en route.	Responder	Must	Implemented
FR-41	The LEAD's device shall share the unit's live location with visible status (live, waiting, denied, error), a pause control, and throttled updates.	Responder	Should	Implemented

ID	Requirement	Actor(s)	Priority	Status
FR-42	An administrator shall be able to force-end a unit's shift and to reconcile unit statuses that no longer match any active mission.	Admin, Super Admin	Should	Implemented
FR-43	A responder shall see incident details and a route map for each active mission, and a list of past missions linking to the incident history.	Responder	Should	Implemented

3.5 Hospital Coordination

ID	Requirement	Actor(s)	Priority	Status
FR-44	When an ambulance mission reaches the scene, the LEAD shall be offered hospitals within 30 km of the incident ranked by distance with the nearest recommended, a searchable list of all hospitals, and a free-text destination as a last resort.	Ambulance Crew	Must	Implemented
FR-45	Hospital selection shall be possible on a map — hover highlights, click selects, an explicit confirmation commits — with the crew's own base and the recommended hospital clearly indicated.	Ambulance Crew	Should	Implemented
FR-46	A hospital transfer shall progress Selected → En Route → Arrived; arrival completes the mission, and the incident resolves when all its missions are complete.	Ambulance Crew	Must	Implemented
FR-47	While en route, the LEAD shall be able to submit patient vitals to a registered hospital; every update is retained.	Ambulance Crew	Must	Implemented
FR-48	Hospital staff shall see incoming transfers for their own facility with a live map of the inbound unit and live vitals.	Hospital Staff	Must	Implemented

3.6 AI-Assisted Decision Support

ID	Requirement	Actor(s)	Priority	Status
FR-49	The system shall classify each incident's priority (Critical, High, Medium, Low) with a confidence value, asynchronously; a failed classification shall never block or alter the incident.	System	Must	Implemented
FR-50	The system shall compare a new incident with open incidents of the same category within 2 km and the previous 120 minutes and surface a suggested duplicate when the model's confidence is at least 0.5.	System	Should	Implemented
FR-51	AI output shall be advisory only; no incident is assigned, cancelled, or closed by AI.	System	Must	Implemented

3.7 Administration & Platform Services

ID	Requirement	Actor(s)	Priority	Status
FR-52	An administrator shall be able to create, edit, and list facilities; a facility's type is fixed after creation, and its registered flag is set automatically when staff or a unit is linked.	Admin, Super Admin	Must	Implemented

ID	Requirement	Actor(s)	Priority	Status
FR-53	An administrator shall be able to seed hospital, fire, and police facilities from OpenStreetMap; duplicates are skipped by coordinates and missing addresses fall back to reverse geocoding. Seeding stops cleanly when the geocoding service rate-limits.	Admin, Super Admin	Should	Implemented
FR-54	An administrator shall be able to retry reverse geocoding for facilities that have no address.	Admin, Super Admin	Should	Implemented
FR-55	An administrator shall be able to create, edit, and delete response units; a unit's home station is limited to facilities of a matching type (an ambulance may be based at an ambulance station or a hospital).	Admin, Super Admin	Must	Implemented
FR-56	An administrator shall be able to search all incidents, filter by status, and moderate them (complete, mark duplicate, retry AI, remove evidence).	Admin, Super Admin	Should	Implemented
FR-57	The interface shall be available in English and Arabic with right-to-left layout, switchable instantly.	All	Must	Implemented
FR-58	Any visitor shall be able to send a message to the support team through a contact form.	All	Could	Implemented
FR-59	Citizens and field responders shall be able to use a native mobile application that talks to the same backend and identity project as the web console.	Citizen, Responder	Should	Implemented

4 Non-Functional Requirements

ID	Category	Requirement
NFR-1	Security — authentication	Every API request is authenticated except registration bootstrap, the email-availability check, the health endpoint, and the WebSocket handshake. Tokens are verified server-side on each call.
NFR-2	Security — authorization	Authorization is enforced in the service layer through method-level role checks and explicit ownership checks. Client-side gating exists for usability only.
NFR-3	Security — data at rest	Chat messages, incident free text, and patient-vitals notes are encrypted with AES-256-GCM using a key supplied through configuration; rows written before encryption remain readable.
NFR-4	Security — media	Evidence and application documents live in a private bucket. Access is only through signed URLs that expire after five minutes and are issued after an authorization check.
NFR-5	Privacy	No public media URLs. Third parties receive only what they need: coordinates for mapping and routing, incident category and text for AI classification.
NFR-6	Responsiveness	Submitting an incident returns without waiting for AI classification or reverse geocoding. Lists update through WebSocket push instead of polling.
NFR-7	Concurrency integrity	Unit assignment takes a pessimistic row lock; incidents and missions carry version columns for optimistic concurrency.
NFR-8	Resilience	Failures of AI, geocoding, or routing degrade gracefully: the incident remains fully usable, routes fall back to a straight line, and rate limiting from a free geocoding service stops a batch instead of repeating failed calls.
NFR-9	Usability	Responsive layouts from phone to desktop, light and dark themes, and confirmation steps for destructive actions.
NFR-10	Localization	English and Arabic with right-to-left layout; direction-sensitive content such as phone numbers and links is kept left-to-right.
NFR-11	Maintainability	Layered backend (controller, service, repository), versioned database migrations, and schema validation at start-up so entity and schema drift fails loudly.
NFR-12	Portability	The full stack (database, backend, web) runs from one Docker Compose file; all environment-specific values are configuration.
NFR-13	Auditability	Privileged identity changes that bypass verification (phone override) are written to an audit log with actor, target, and before/after values.
NFR-14	Interoperability	REST over JSON and STOMP over WebSocket; the mobile and web clients share one API contract.

5 User Stories

5.1 Identity & Access Management

ID	User story	Requirement
US-1	As a citizen, I want to register with just my email and a link, so that I never have to invent a password before I know the account is mine.	FR-1, FR-3
US-2	As a citizen, I want to finish my profile and set a password after confirming my email, so that I can sign in normally next time.	FR-2
US-3	As a citizen, I want to be told early if my email already has an account, so that I sign in instead of registering twice.	FR-4
US-4	As an administrator, I want to create employee accounts for any role, so that staff can access only their own console.	FR-6, FR-7
US-5	As an administrator, I want to search, disable, and re-role accounts, so that access always matches someone's current job.	FR-8, FR-9, FR-10
US-6	As a Super Admin, I want to override a phone number and reconcile Firebase with the database, so that I can repair accounts without bypassing the audit trail.	FR-11, FR-18
US-7	As a user, I want to manage my own profile, email, phone, and password, so that my details stay current.	FR-14
US-8	As a citizen, I want to apply to become a First Responder with my certifications, so that I can help respond.	FR-16
US-9	As an administrator, I want to review applications and their documents, so that only vetted people become responders.	FR-17

5.2 Emergency Reporting

ID	User story	Requirement
US-10	As a citizen, I want to report an emergency in two taps and confirm where it is, so that help is sent to the right place.	FR-19
US-11	As a citizen, I want to attach photos, video, or audio, so that responders understand what they are heading into.	FR-21
US-12	As a citizen, I want to correct my report until a unit is assigned, so that dispatch works from accurate information.	FR-22
US-13	As a citizen, I want to see my active and past incidents, so that I know what happened.	FR-23
US-14	As a citizen, I want to talk to whoever is responding, so that I can answer their questions.	FR-26
US-15	As a citizen, I want my messages and evidence protected, so that sensitive details stay private.	FR-24, FR-27

5.3 Dispatch Operations

ID	User story	Requirement
US-16	As a dispatcher, I want a live queue with AI priority, so that I handle the most urgent incident first.	FR-28, FR-49
US-17	As a dispatcher, I want to pick a unit on a map sorted by distance, so that the closest suitable unit goes.	FR-29, FR-34

ID	User story	Requirement
US-18	As a dispatcher, I want two of us to be unable to send the same unit twice, so that dispatch never collides.	FR-30
US-19	As a dispatcher, I want to cancel or complete assignments, so that I can correct mistakes and close incidents.	FR-31, FR-32
US-20	As a dispatcher, I want duplicate reports flagged, so that one event is not worked twice.	FR-35, FR-50
US-21	As a Super Admin, I want to delete erroneous incidents, so that test data does not pollute records.	FR-33

5.4 Field Response

ID	User story	Requirement
US-22	As a responder, I want to start or join a shift for my unit, so that I only receive work when I am on duty.	FR-37, FR-38
US-23	As a unit LEAD, I want to accept or reject an offered mission, so that I stay in control of my crew.	FR-39
US-24	As a unit LEAD, I want to report en route, arrived, and completed, and to withdraw if I cannot proceed, so that dispatch always knows the truth.	FR-40
US-25	As a responder, I want a live route to the incident and my location shared automatically, so that dispatch sees me move.	FR-41, FR-43
US-26	As an administrator, I want to end a stuck shift and repair stale unit statuses, so that units are never lost to bad state.	FR-42

5.5 Hospital Coordination

ID	User story	Requirement
US-27	As an ambulance LEAD, I want recommended and searchable hospitals on a map, so that I choose a destination quickly.	FR-44, FR-45
US-28	As an ambulance LEAD, I want to send live vitals while en route, so that the hospital is ready on arrival.	FR-47
US-29	As hospital staff, I want to see inbound ambulances and their vitals live, so that we can prepare.	FR-48
US-30	As an ambulance LEAD, I want arrival to finish my mission, so that I do not close it twice.	FR-46

5.6 AI-Assisted Decision Support

ID	User story	Requirement
US-31	As a dispatcher, I want AI priority and duplicate suggestions that I can accept or dismiss, so that AI helps without deciding.	FR-49, FR-50, FR-51
US-32	As a dispatcher, I want to retry a failed classification, so that a transient outage does not leave an incident unranked.	FR-36

5.7 Administration & Platform

ID	User story	Requirement
US-33	As an administrator, I want to manage facilities and seed them from OpenStreetMap, so that the map reflects real places.	FR-52, FR-53, FR-54
US-34	As an administrator, I want units tied to appropriate home stations, so that data stays consistent.	FR-55
US-35	As an administrator, I want to search and moderate every incident, so that I can audit what happened.	FR-56
US-36	As any user, I want the app in Arabic or English, so that I can use it in my own language.	FR-57
US-37	As a visitor, I want to contact the team, so that I can report a problem.	FR-58
US-38	As a citizen or responder in the field, I want a mobile app, so that I am not tied to a desk.	FR-59

6 Use Case Analysis

This section describes the system's use cases, grouped into seven bounded contexts. Each group opens with a use case diagram, followed by detailed descriptions in a common template: actor, platform, preconditions, postconditions, the main flow, and any alternate or exception flows.

Diagram notation. Human actors are shown as person icons and automated actors as «system» boxes. <<include>> marks behaviour always performed as part of the base use case; <<extend>> marks optional, conditional behaviour; <<triggers>> (a practical extension to UML) marks an automated actor initiating a use case as a consequence of another event. Use cases shaded blue are restricted to administrative roles.

Platform. Use cases are available on the Web console, on the Mobile companion app, or both, as stated in each description. Mobile covers the Citizen and Responder roles.

Group	Use cases	Figure
6.1 Identity & Access Management	UC-1 to UC-5, UC-30, UC-31	Figure 6.1
6.2 Emergency Reporting	UC-6, UC-17, UC-18, UC-19	Figure 6.2
6.3 Dispatch Operations	UC-7, UC-13 to UC-16, UC-29	Figure 6.3
6.4 Field Response & Shift Management	UC-8, UC-10, UC-11, UC-12	Figure 6.4
6.5 Hospital Coordination	UC-9, UC-20	Figure 6.5
6.6 Facility & System Administration	UC-21 to UC-26	Figure 6.6
6.7 General & Support Services	UC-27, UC-28	Figure 6.7

6.1 Identity & Access Management

Registration, sign-in, profile, role, and account administration. Blue use cases are restricted to Admin and Super Admin.

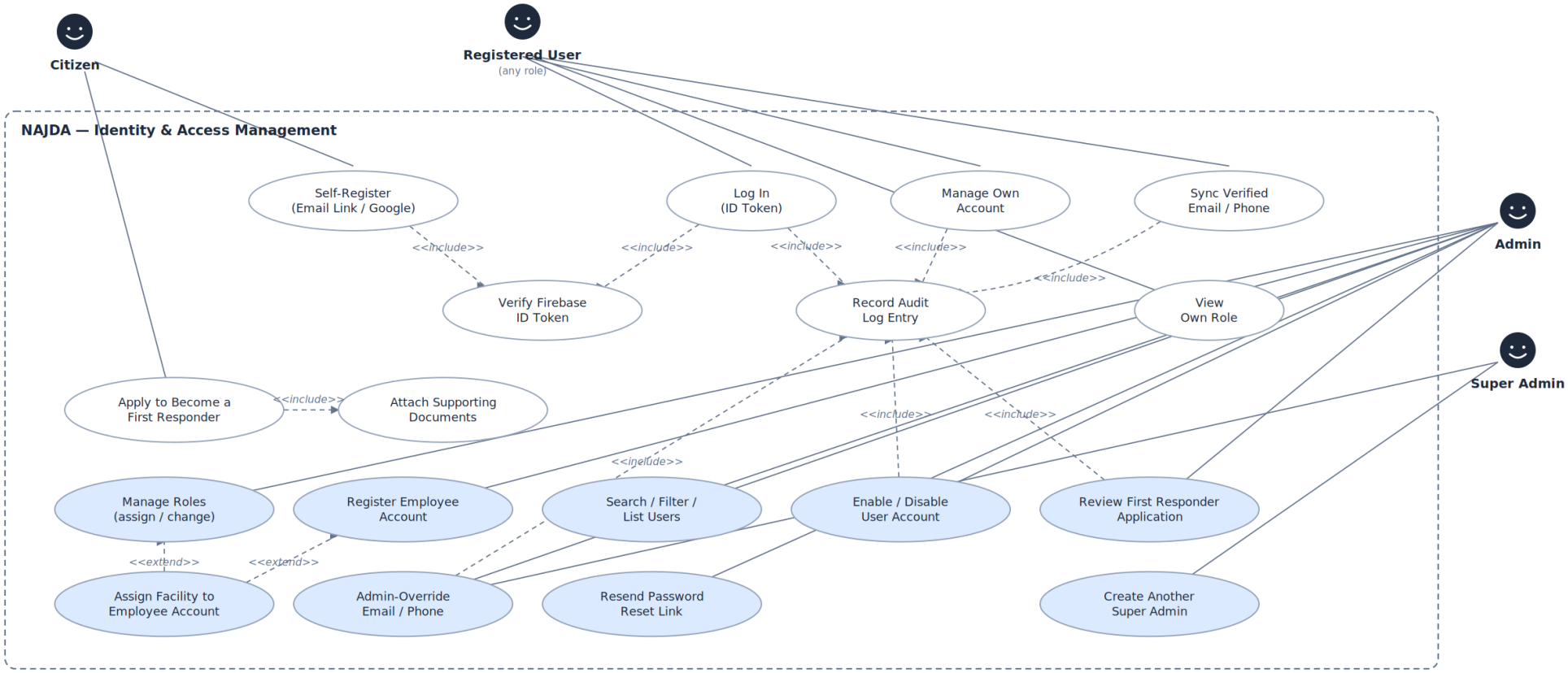


Figure 6.1 — Identity & Access Management use cases

UC-1 — Citizen Self-Registration

Field	Detail
Actor	Citizen
Platform	Web, Mobile
Preconditions	The person has no NAJDA account and has accepted the Terms & Conditions and Privacy Policy.
Postconditions (success)	A citizen account exists with a confirmed email, a completed profile, and a password.
Postconditions (failure)	No database record exists for an unconfirmed email address.

Main flow:

1. The citizen enters an email address and accepts the terms, or chooses Continue with Google.
2. The system sends a passwordless sign-in link to the address. No account exists yet.
3. The citizen opens the link. On the same browser the email is recalled automatically; on another device the citizen re-enters it once.
4. Firebase confirms the link, signs the citizen in, and marks the email verified in the same step.
5. The system creates the NAJDA user record with blank names and `profileCompleted = false`.
6. The profile gate routes the citizen to profile completion: first and last name, phone number (+20, validated), address, gender, and a password that meets the strength rules.
7. The system saves the profile; with a phone number, address, gender, and a password provider present the profile is complete and the citizen enters the app.

Alternate flow — Google:

1. At step 1 the citizen chooses Google; Google asserts the email.
2. The system creates the user record immediately with the Google name; the citizen still completes phone, address, and gender (no password is required).

Alternate flow — resuming an abandoned registration:

1. If the citizen requests a new link for an email that already has a record, the system treats the existing record as success and continues at step 6.

Alternate flow — email already registered:

1. As the citizen types, the form shows whether the address already has an account and offers sign-in instead.

Exception flow — invalid or expired link:

1. The system shows an error and the citizen requests a new link.

UC-2 — Admin-Provisioned Employee Registration

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	The caller holds ADMIN or SUPER_ADMIN. Creating a SUPER_ADMIN additionally requires the caller to be a Super Admin.
Postconditions (success)	A Firebase account and a matching NAJDA user exist with the requested role (and facility, where applicable), and a one-time password-reset link has been generated.
Postconditions (failure)	No orphaned Firebase account is left behind — if the database write fails, the Firebase account is deleted.

Main flow:

1. The admin enters the employee's name, email, phone, address, gender, and role.

2. If the role links to a facility, the admin selects a facility of the matching type (required for Hospital Staff). The facility is marked registered.
3. The system creates a Firebase account with a generated temporary password and the matching user record, and sets the role claim.
4. For a First Responder the system also creates a personal response unit.
5. The system generates a password-reset link and returns it. It is not emailed; the admin shares it so the employee can set their own password.

Alternate flow — requested role is SUPER_ADMIN and caller is not:

1. The request is rejected with access denied before any account is created.

UC-3 — Enable / Disable a User Account

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	The target is not a Super Admin unless the caller is also a Super Admin; the last enabled Super Admin cannot be disabled.
Postconditions (success)	The account's enabled flag is flipped; a disabled account can no longer authenticate.

Main flow:

1. The admin opens the user and toggles Enable / Disable.
2. The system updates the flag and timestamp; the change takes effect on the user's next request.

UC-4 — Assign or Change a User's Role

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	Same Super Admin protection as UC-3.
Postconditions (success)	The user's role (and facility, where relevant) is updated and the role claim re-synchronized.

Main flow:

1. The admin selects a new role.
2. If the role links to a facility the admin selects one of the matching type; if it does not, any existing link is cleared.
3. The system saves the change and re-syncs the role claim so authorization reflects it immediately.

UC-5 — First Responder Certification Application

Field	Detail
Actor	Citizen (applicant); Admin, Super Admin (reviewer)
Platform	Web
Preconditions	The applicant's email and phone are verified and they have no pending application.
Postconditions (success)	Approved: the account's role is FIRST_RESPONDER with a personal unit. Rejected: the applicant may apply again.

Main flow:

1. The citizen writes a motivation and attaches at least one document (image, PDF, or Word).

2. The system stores the application and every document in a single request; an application never exists without its documents.
3. An admin opens the pending application and views each document through a short-lived signed link.
4. The admin approves; the system sets the role to First Responder and creates the applicant's personal unit.

Alternate flow — rejected:

1. At step 4 the admin rejects with an optional reason; the applicant sees it and may re-apply.

Alternate flow — document removed:

1. An admin may remove a document, which deletes the file from storage and the record.

UC-30 — Administer User Accounts

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	The caller holds ADMIN or SUPER_ADMIN.
Postconditions (success)	The requested change is applied; privileged phone overrides are audit-logged.

Main flow:

1. The admin searches and filters the user list by name, email, phone, role, and enabled state.
2. The admin opens a user and may edit profile fields, override the email (the identity record is updated and re-verification is required), generate a password-reset link, enable or disable the account (UC-3), or change the role (UC-4).
3. A Super Admin may also override the phone number; the system writes an audit entry with the previous and new values, and refuses a number that is already verified on another account.

UC-31 — Manage Own Account

Field	Detail
Actor	Any authenticated user
Platform	Web, Mobile
Preconditions	The user is signed in.
Postconditions (success)	The user's own profile, email, phone, or password reflects the change.

Main flow:

1. The user edits name, address, or gender and saves.
2. To change email, the user enters the new address; a confirmation link is sent and the email changes only when the link is opened.
3. To change phone, the user saves the number (unverified), then verifies it with a one-time code; a number already verified elsewhere is refused.
4. To change password, the user re-enters the current password and the new one (accounts that sign in with Google have no password to change).

6.2 Emergency Reporting

The citizen's reporting experience, with the automated AI and geocoding services that act on a new report.

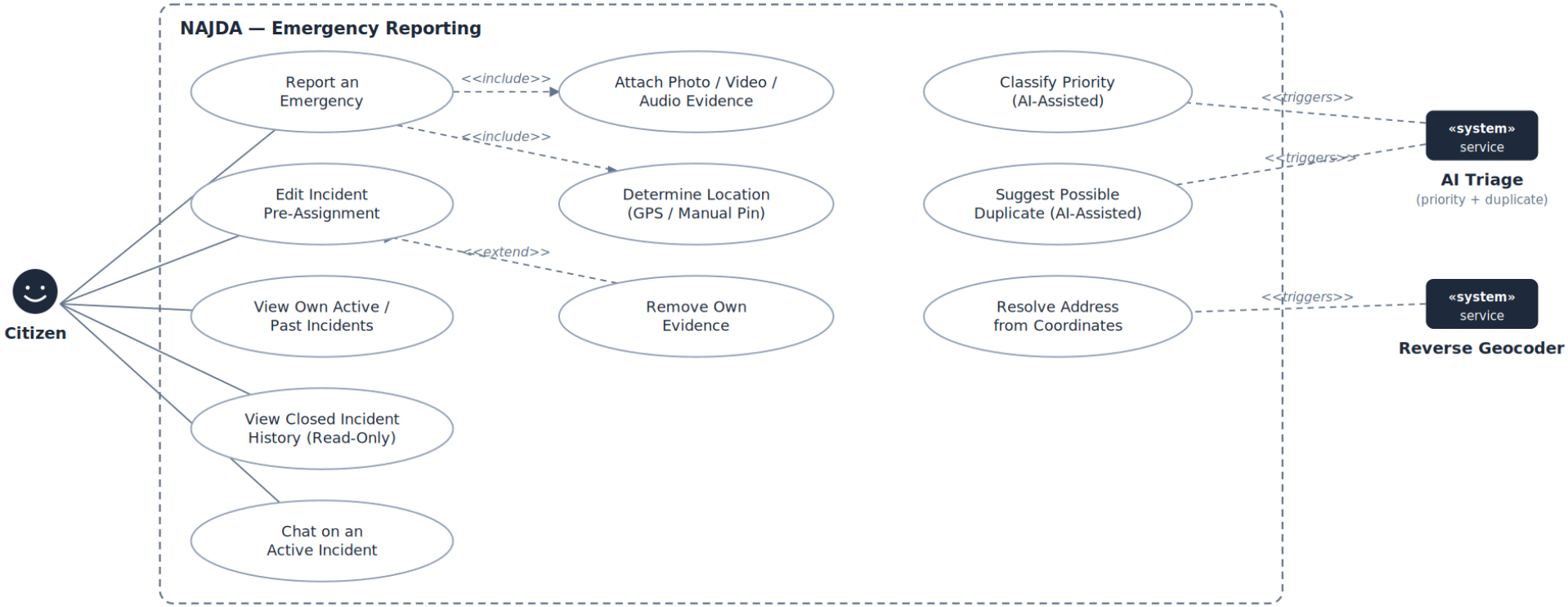


Figure 6.2 — Emergency Reporting use cases

UC-6 — Report an Emergency

Field	Detail
Actor	Citizen
Platform	Web, Mobile (evidence attachment: Web)
Preconditions	The citizen is signed in, has a completed profile, and has no active incident.
Postconditions (success)	An incident exists in NEW status; AI priority, duplicate suggestion, and address populate asynchronously.
Postconditions (failure)	No incident is created.

Main flow:

1. The citizen presses the SOS control and chooses a category: Medical, Fire, or Police.
2. The citizen optionally describes the situation and sets the injured count (default zero).
3. The location comes from device GPS or a manually placed pin; a point outside Egypt is rejected.
4. (Web) The citizen may attach photo, video, or audio. Each file is uploaded straight to private storage through a short-lived signed URL, then registered with the incident once the system confirms the file exists.
5. The citizen submits. The system creates the incident (NEW) and pushes it to every connected dispatcher.
6. In the background the system requests AI priority classification, an AI duplicate check, and a reverse-geocoded address; none of these delays the response.

Alternate flow — partial evidence failure:

1. If some files fail to upload but the incident exists, the citizen is told how many failed; the incident stays valid and dispatchable.

Exception flow — active incident exists:

1. The system rejects the submission with a conflict; the citizen is shown their existing incident.

UC-17 — Edit an Incident Before Assignment

Field	Detail
Actor	Citizen
Platform	Web (mobile: description and injured count)
Preconditions	The incident belongs to the citizen and is NEW, AI_PROCESSED, or DISPATCHER_REVIEW.
Postconditions (success)	The incident reflects the edit and connected dispatchers see it immediately.
Postconditions (failure)	Editing is refused once a unit is assigned.

Main flow:

1. The citizen edits the description, changes the injured count, or adds or removes evidence.
2. The system saves and pushes the update.

Exception flow — already assigned:

1. The system rejects the edit with a conflict, because a responder or dispatcher may already be acting on what was reported.

UC-18 — View Closed Incident History

Field	Detail
Actor	Citizen; any user who participated in the incident (including a responder whose mission has completed)

Field	Detail
Platform	Web
Preconditions	The incident is RESOLVED or CANCELLED.
Postconditions (success)	None — read-only.
Postconditions (failure)	A user who never took part receives access denied.

Main flow:

1. The user opens the incident from their history.
2. The system shows category, status, map, address, original message, evidence, and the full chat transcript, all read-only.

UC-19 — Chat on an Incident

Field	Detail
Actor	Citizen, Dispatcher, Admin, responder on an active mission
Platform	Web
Preconditions	The incident is not closed. Sending is limited to those actors; anyone who ever participated may read.
Postconditions (success)	The message is stored encrypted and delivered live to every connected participant.

Main flow:

1. The actor types a message and sends.
2. The system verifies the sender may write to this incident, encrypts and stores the message, and pushes it to subscribers.

6.3 Dispatch Operations

Admin and Super Admin hold every Dispatcher capability shown here; Delete an Incident is restricted to Super Admin.

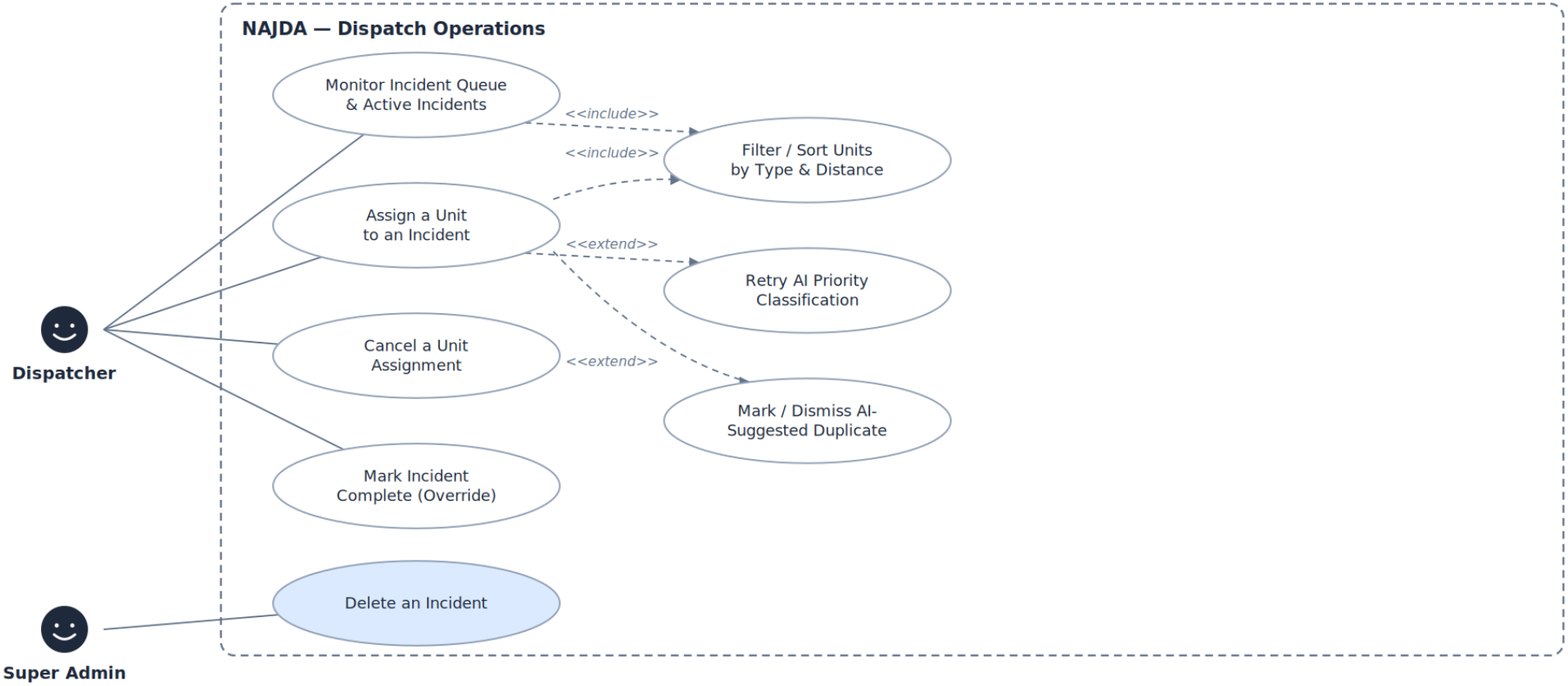


Figure 6.3 — Dispatch Operations use cases

UC-7 — Assign a Unit to an Incident

Field	Detail
Actor	Dispatcher, Admin, Super Admin
Platform	Web
Preconditions	The incident is not RESOLVED or CANCELLED.
Postconditions (success)	A mission exists in OFFERED status, the unit is BUSY, and the incident is ASSIGNED.

Main flow:

1. The dispatcher monitors the live queue and opens an incident; the live map shows it with every relevant unit.
2. The dispatcher filters units by type; the list is sorted nearest-first using each unit's live position or its home facility, with an estimated arrival time.
3. The dispatcher selects a unit on the map or in the list and confirms.
4. The system locks the unit row, verifies it is still AVAILABLE, creates the mission (OFFERED), marks the unit BUSY, and snapshots the crew currently on shift as mission participants.
5. On the first assignment the incident becomes ASSIGNED; additional units on the same incident do not repeat the transition.
6. The responding LEAD is notified live.

Exception flow — unit no longer available:

1. If another dispatcher claimed the unit first, the second attempt fails cleanly with a conflict.

UC-29 — Cancel a Unit Assignment

Field	Detail
Actor	Dispatcher, Admin, Super Admin
Platform	Web
Preconditions	The mission is not COMPLETED, CANCELLED, or REJECTED.
Postconditions (success)	The mission is CANCELLED, the unit is AVAILABLE, and the incident returns to DISPATCHER_REVIEW if no other mission is still active.

Main flow:

1. The dispatcher chooses Cancel assignment on a mission and confirms.
2. The system cancels the mission, frees the unit, and reopens the incident for reassignment if nothing else is working it.

UC-13 — Retry AI Priority Classification

Field	Detail
Actor	Dispatcher, Admin, Super Admin
Platform	Web
Preconditions	The incident has no AI priority.
Postconditions (success)	Priority and confidence populate, or remain unset if the retry fails.

Main flow:

1. The dispatcher chooses Retry AI.
2. The system re-runs classification in the background and pushes the result when it completes.

UC-14 — Mark or Dismiss a Duplicate

Field	Detail
Actor	Dispatcher, Admin, Super Admin
Platform	Web
Preconditions	The incident carries an AI-suggested duplicate, or the dispatcher chooses to mark one manually.
Postconditions (success)	Confirmed: the incident is cancelled and linked to the original. Dismissed: the suggestion is cleared.

Main flow:

1. The dispatcher reviews the suggestion with its confidence.
2. On Confirm the system links the incident to the original as a duplicate and cancels it, recording reason, actor, and time.

Alternate flow — dismiss:

1. The dispatcher chooses Not a duplicate; the system clears the suggestion and the incident continues.

Alternate flow — manual:

1. The dispatcher picks any other open incident from the queue as the original.

UC-15 — Mark an Incident Complete (Override)

Field	Detail
Actor	Dispatcher, Admin, Super Admin
Platform	Web
Preconditions	The incident is not RESOLVED or CANCELLED.
Postconditions (success)	The incident is RESOLVED, every active mission is COMPLETED, and every involved unit is AVAILABLE.

Main flow:

1. The dispatcher chooses Mark Incident Complete and confirms the warning.
2. The system completes every active mission, frees every unit, and resolves the incident.

UC-16 — Delete an Incident

Field	Detail
Actor	Super Admin
Platform	Web
Preconditions	None beyond the role — the action is unconditional and irreversible.
Postconditions (success)	The incident, its missions, chat, and hospital transfers are removed and any unit still busy on it is freed.

Main flow:

1. The Super Admin chooses Delete incident and confirms the explicit warning.
2. The system frees any busy unit, removes dependent records, then the incident.

6.4 Field Response & Shift Management

The responder's duty cycle and mission progression. Force-ending a shift is an administrative recovery action.

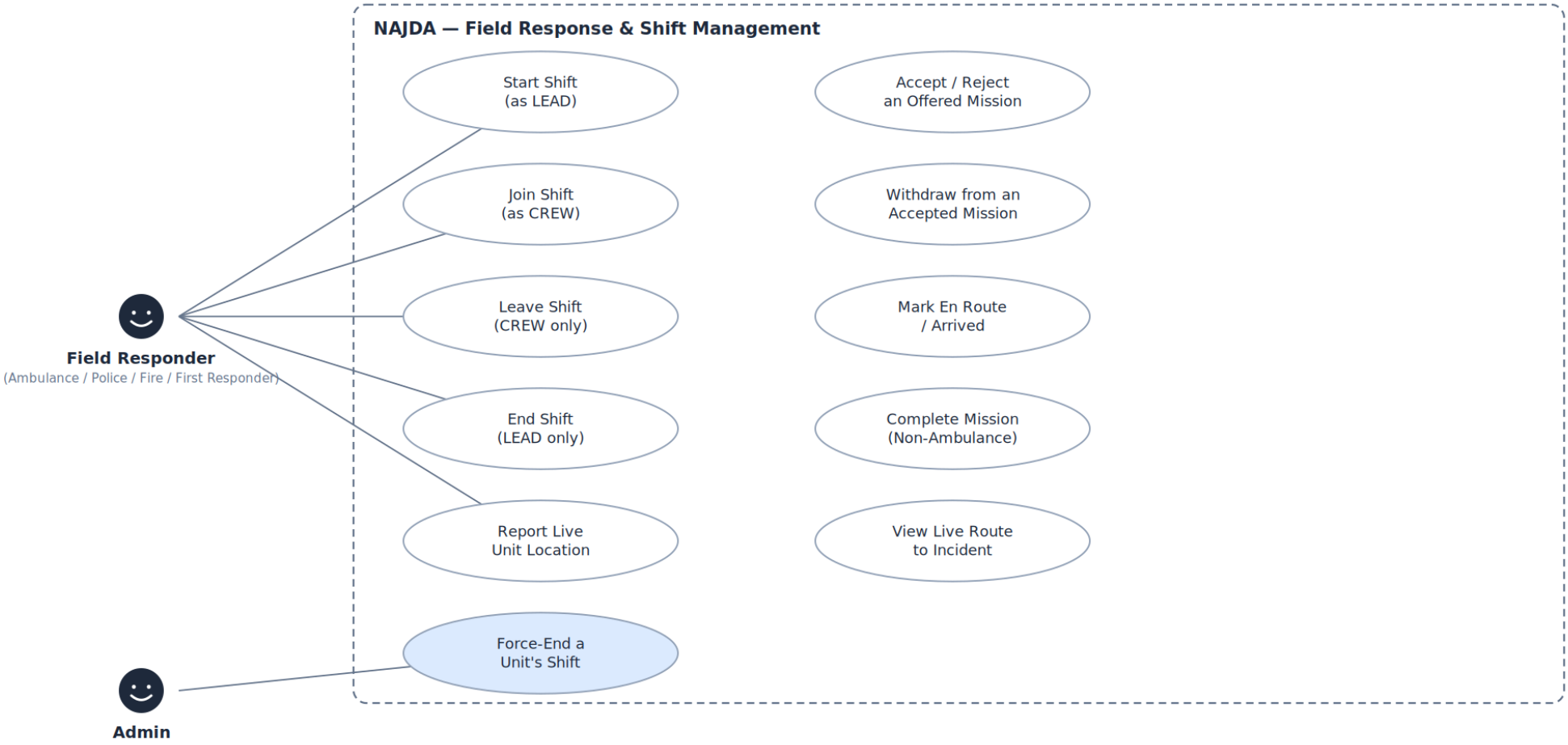


Figure 6.4 — Field Response & Shift Management use cases

UC-10 — Start, Join, Leave, and End a Shift

Field	Detail
Actor	Ambulance Crew, Police, Firefighter, First Responder
Platform	Web, Mobile
Preconditions	The employee has no active shift. Roles tied to a facility may only use units at their own facility; Ambulance Crew and First Responder are exempt from that restriction. A First Responder sees only their own unit.
Postconditions (success)	The shift record reflects the change and the unit's availability follows it.

Main flow:

1. To start, the employee selects a unit that has no LEAD; the system opens a LEAD shift and the unit becomes available.
2. To join, the employee selects a unit that already has a LEAD and joins as CREW.
3. A CREW member may leave at any time.
4. The LEAD ends the shift, which closes every open shift on the unit.

Exception flow — ineligible:

1. If the unit belongs to another facility, or the employee already has an active shift, the system refuses with an explanatory message.

UC-8 — Work a Mission

Field	Detail
Actor	Unit LEAD (CREW members see the mission but do not act on it)
Platform	Web, Mobile
Preconditions	The employee is the active LEAD of the unit and a mission is OFFERED to it.
Postconditions (success)	The mission reaches COMPLETED (or CANCELLED / REJECTED); the unit is AVAILABLE again.

Main flow:

1. The LEAD is notified live and accepts the offered mission; it becomes ACCEPTED.
2. The LEAD marks the unit en route; the live route map and shared location guide the way.
3. The LEAD marks arrival on scene (ARRIVED).
4. For non-ambulance units the LEAD completes the mission; the unit is freed, and when every mission on the incident is complete the incident resolves.
5. For an ambulance the LEAD continues with the hospital handoff (UC-9).

Alternate flow — reject:

1. At step 1 the LEAD rejects with a reason; the mission becomes REJECTED and the unit is freed so the dispatcher can assign another.

Alternate flow — withdraw:

1. While the mission is ACCEPTED or EN_ROUTE, the LEAD may withdraw (with confirmation); the mission is CANCELLED, the unit is freed, and the incident is reopened if nothing else is working it.
2. After arrival the LEAD must ask a dispatcher to cancel instead.

Exception flow — not the LEAD:

1. Any mission action by someone who is not the unit's active LEAD is refused by the server.

UC-12 — Report Live Unit Location

Field	Detail
Actor	Unit LEAD
Platform	Web, Mobile
Preconditions	The employee is the active LEAD and has granted location permission.
Postconditions (success)	The unit's position is stored and every dispatch map updates live.

Main flow:

1. While on shift the device watches its position and sends an update at most every 15 seconds.
2. The shift card shows the sharing state: live with the last update time, waiting for GPS, permission denied, unsupported, or error.
3. The LEAD may pause and resume sharing.

Alternate flow — no live position:

1. Dispatch views fall back to the unit's home facility location and label it as such.

UC-11 — Force-End a Unit's Shift

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	The unit has an active LEAD.
Postconditions (success)	Every open shift on the unit is closed and the unit is OFFLINE.

Main flow:

1. The admin opens the unit list and chooses Force End Shift on a unit with an active LEAD.
2. The system closes every open shift on that unit.

6.5 Hospital Coordination

Destination selection and live patient vitals for ambulance missions, and the hospital's view of the inbound patient.

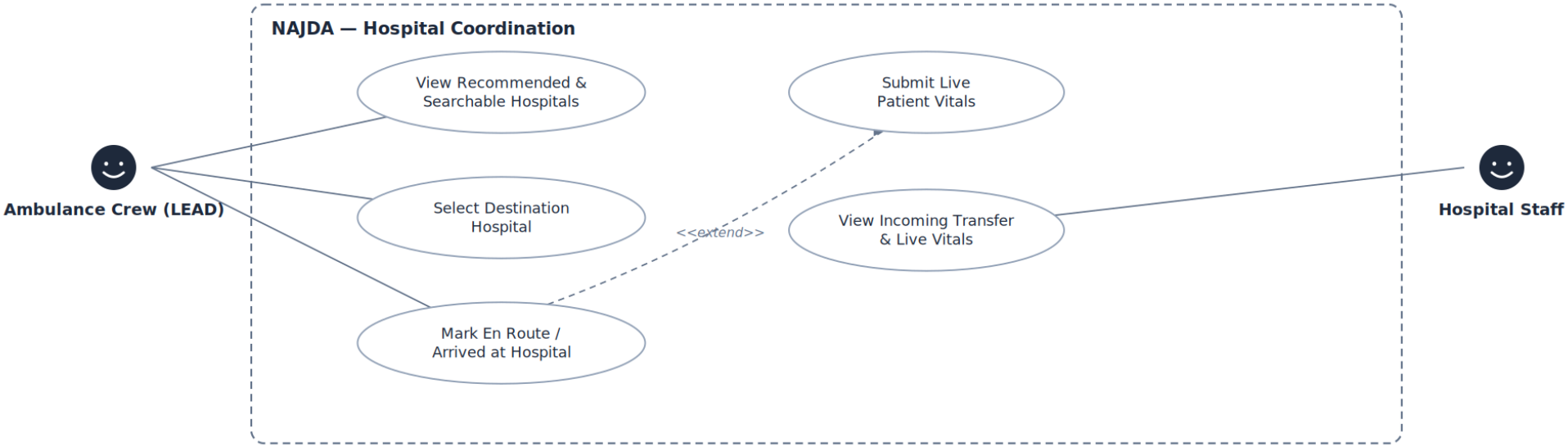


Figure 6.5 — Hospital Coordination use cases

UC-9 — Ambulance Hospital Handoff

Field	Detail
Actor	Ambulance Crew (LEAD)
Platform	Web
Preconditions	The ambulance mission is ARRIVED and has no transfer yet.
Postconditions (success)	The transfer is ARRIVED, the mission is COMPLETED, the unit is AVAILABLE, and the incident resolves when all its missions are complete.

Main flow:

1. The system lists hospitals within 30 km of the incident ranked by distance from the incident, with the nearest recommended, and the full list is searchable. The map shows the incident, every hospital, the recommended hospital, and the crew's own base.
2. The LEAD hovers to highlight and clicks to select a hospital, then confirms; the transfer is SELECTED.
3. The LEAD marks the transfer en route; the route map guides the crew to the hospital.
4. While en route the LEAD submits patient vitals (UC-20 shows them live to the hospital).
5. The LEAD marks arrival; the mission completes and the unit is freed.

Alternate flow — destination not in the system:

1. The LEAD enters a hospital name as free text; the transfer proceeds, but live vitals are not accepted because there is no registered hospital to receive them.

Exception flow — hospital not registered:

1. Vitals submitted for a hospital with no linked staff are refused with an explanatory message.

UC-20 — View Incoming Transfer and Live Vitals

Field	Detail
Actor	Hospital Staff
Platform	Web
Preconditions	The account is linked to a hospital.
Postconditions (success)	None — read-only.
Postconditions (failure)	An account with no facility link sees a message to contact an administrator.

Main flow:

1. The staff member sees incoming transfers for their hospital and a list of arrived patients.
2. Selecting a transfer shows a map of the pickup location and the inbound unit.
3. The vitals panel updates live as the crew submits readings.

6.6 Facility & System Administration

Facility and unit management, data seeding, and reconciliation tools. Blue use cases are restricted to Super Admin.

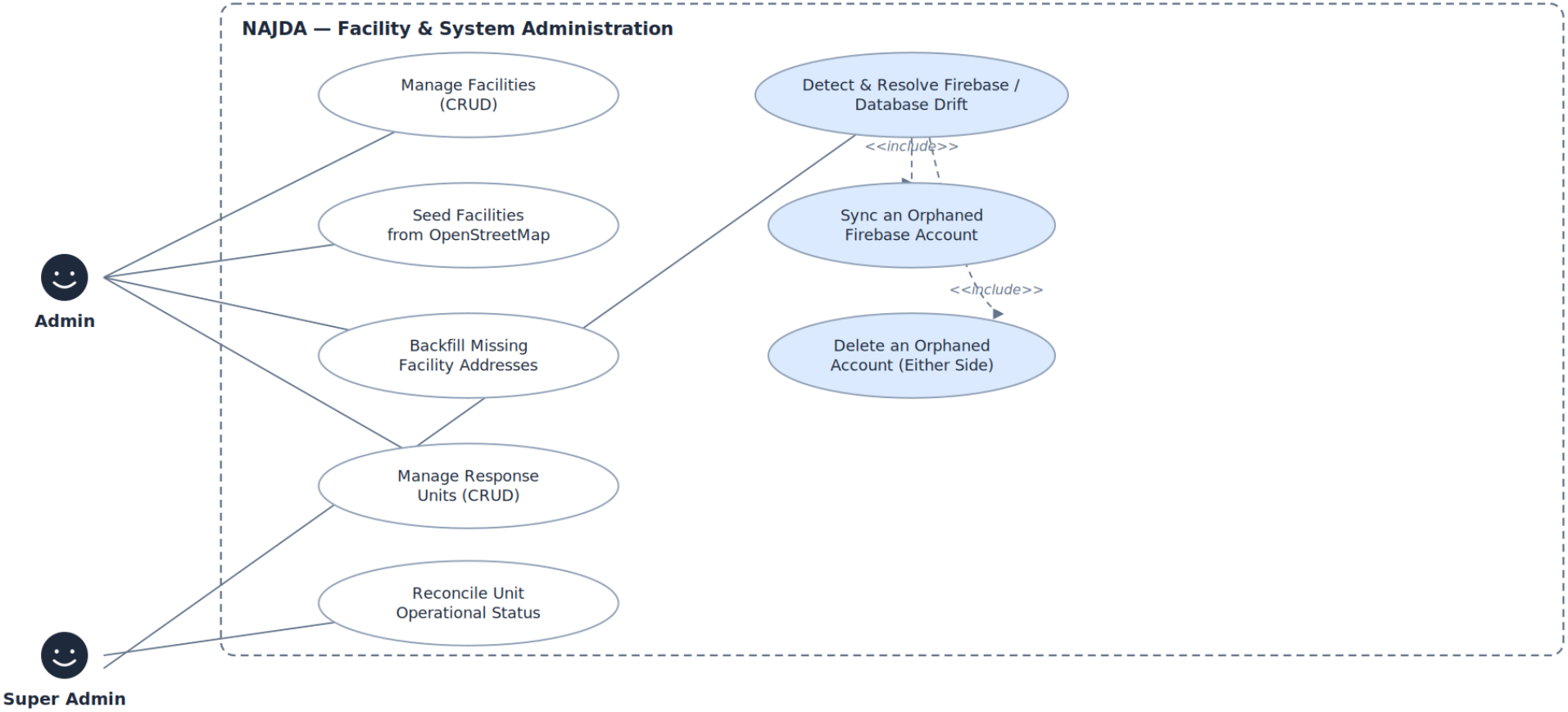


Figure 6.6 — Facility & System Administration use cases

UC-21 — Manage Facilities

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	The caller holds ADMIN or SUPER_ADMIN.
Postconditions (success)	The facility is saved.

Main flow:

1. The admin lists facilities with search and a type filter.
2. The admin creates a facility (type, name, address, coordinates) or edits one; the type cannot change after creation.
3. The registered flag is never set by hand: it turns on when staff or a unit is linked.

UC-22 — Seed Facilities from OpenStreetMap

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	Seeding is available for Hospital, Fire, and Police facilities (ambulance stations are added manually).
Postconditions (success)	New facilities exist, unregistered, with addresses where available.

Main flow:

1. The admin chooses a facility type.
2. The system queries OpenStreetMap for that type across Egypt through a pool of mirrors, skipping unnamed records and any record whose coordinates already exist.
3. For each new facility the address comes from OSM tags or, if absent, from reverse geocoding paced to the service's rate limit.

Alternate flow — rate limited:

1. If the geocoding service rate-limits, the run stops cleanly with a single log message; the admin can re-run later.

UC-23 — Backfill Missing Facility Addresses

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	At least one facility has no address.
Postconditions (success)	Addresses are filled where the geocoder succeeds.

Main flow:

1. The admin chooses Fill missing addresses.
2. The system retries reverse geocoding for each facility with no address, paced to the service limit, and stops on a rate-limit response.

UC-24 — Manage Response Units

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	The caller holds ADMIN or SUPER_ADMIN.
Postconditions (success)	The unit is saved or removed.

Main flow:

1. The admin creates or edits a unit: type, plate number, and home station.
2. The home station picker offers only facilities of a matching type — fire trucks at fire stations, police cars at police stations, ambulances at ambulance stations or hospitals; a First Responder unit has no station.
3. The admin may delete a unit; a unit still in use is refused with a conflict.

UC-25 — Reconcile Unit Operational Status

Field	Detail
Actor	Admin, Super Admin
Platform	Web
Preconditions	None.
Postconditions (success)	No unit remains BUSY without an active mission.

Main flow:

1. The admin chooses Refresh on the unit list.
2. The system finds every BUSY unit with no active mission and returns it to AVAILABLE. Units that are OFFLINE are never touched.

UC-26 — Detect and Resolve Firebase / Database Drift

Field	Detail
Actor	Super Admin
Platform	Web
Preconditions	The caller is a Super Admin. Intended for development setups where several environments share one Firebase project.
Postconditions (success)	Orphans are synced or removed.

Main flow:

1. The Super Admin runs the check; the system compares every Firebase account with the database.
2. For an account in Firebase only, the Super Admin may create a profile: the name is guessed from the email address, a verified Firebase phone number is carried over, and the profile is left incomplete.
3. The Super Admin may instead delete an orphan on either side; the system refuses if the other side turns out to exist.

6.7 General & Support Services

Capabilities available to every user regardless of role.

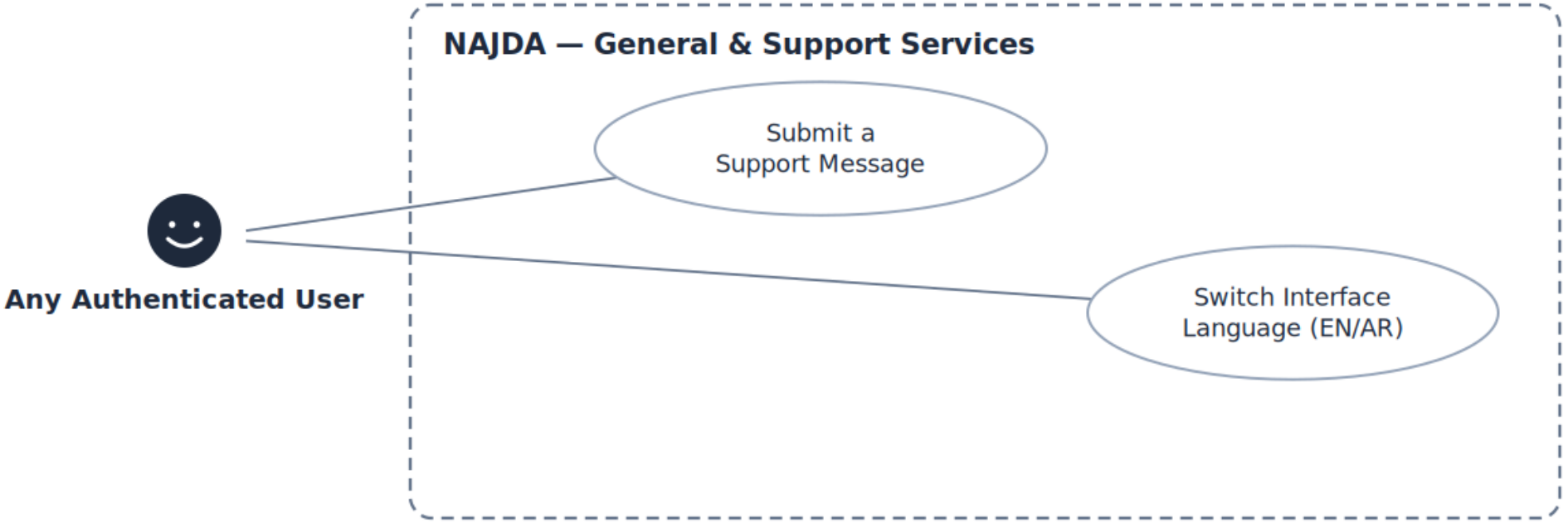


Figure 6.7 — General & Support Services use cases

UC-27 — Submit a Support Message

Field	Detail
Actor	Any user
Platform	Web, Mobile
Preconditions	None.
Postconditions (success)	The message is delivered to the support team by email.

Main flow:

1. The user opens Support and enters a name, email, and message.
2. The system sends it through the email service and confirms delivery.

UC-28 — Switch Interface Language

Field	Detail
Actor	Any user
Platform	Web, Mobile
Preconditions	None.
Postconditions (success)	The interface is shown in the chosen language and direction, remembered for next time.

Main flow:

1. The user chooses English or Arabic.
2. The interface switches instantly, without reloading; Arabic switches to right-to-left reading direction while numbers, phone numbers, and links stay left-to-right.

7 System Architecture and Design

7.1 Architecture Overview

NAJDA is a three-tier system with managed services for identity, object storage, and AI. Two clients — a Next.js web console and an Expo mobile application — talk to a single Spring Boot API over REST and a STOMP WebSocket. The API persists to PostgreSQL and integrates with external services for identity, media, AI classification, geocoding, and routing.

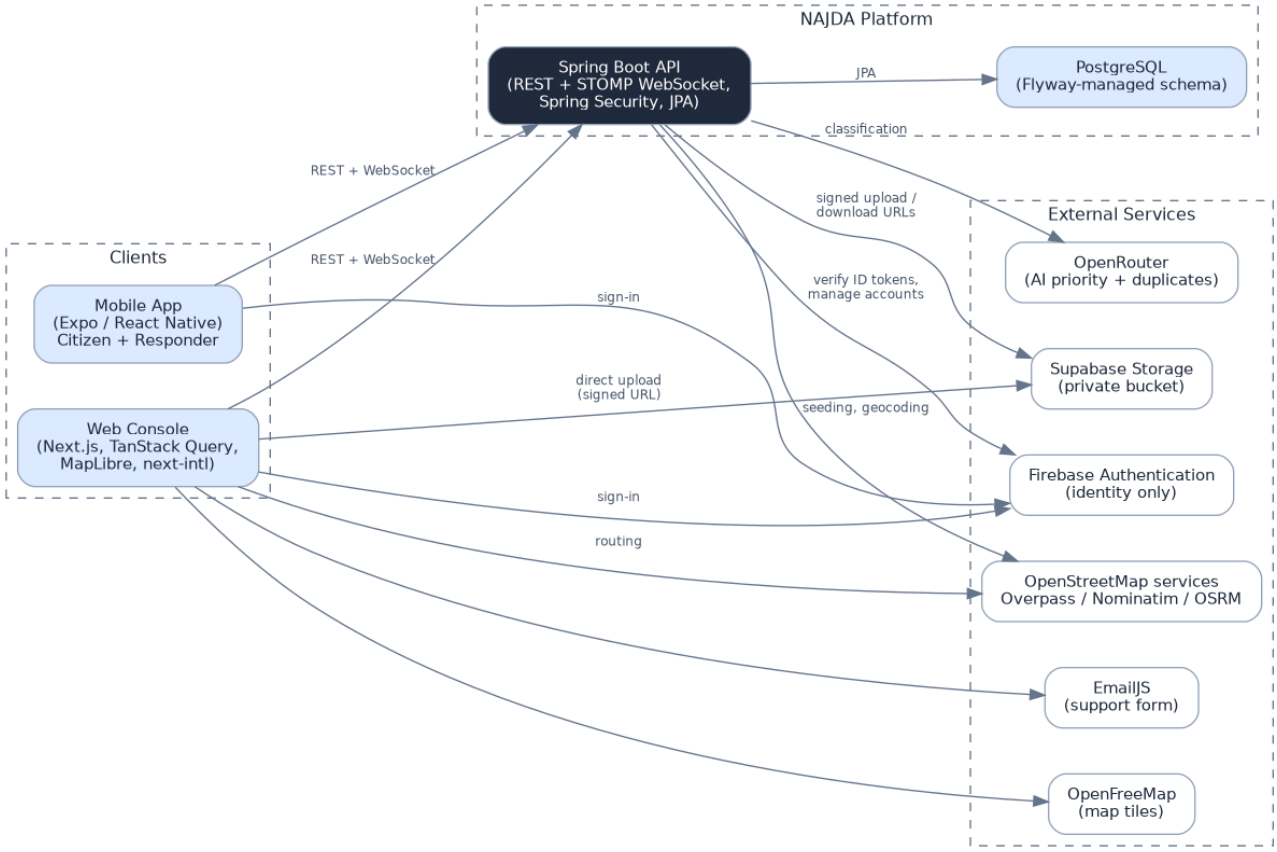


Figure 7.1 — System architecture and external services

Architectural principles

- **Identity is delegated; authority is local.** Firebase proves who someone is. The NAJDA database alone decides what they may do: roles, facility links, and everything that has happened.
- **The backend is the security boundary.** Authorization is enforced in the service layer; client-side gating only shapes the interface.
- **Live by default.** State changes are pushed over WebSocket so consoles react instead of polling.
- **Asynchronous and advisory AI.** Classification, duplicate checks, and geocoding run off the request path, and a failure never affects the incident.
- **Private by default.** Media is never public; sensitive text is encrypted before it reaches the database.
- **One source of truth per fact.** Facilities are one entity with a type; a shift record is the sole evidence a unit has an operator.

Real-time channels

The API exposes a STOMP endpoint at `/ws` backed by Spring's in-memory broker. Clients subscribe to topic channels and refresh the affected data when a message arrives.

Topic	Published when	Consumers
/topic/incidents	An incident is created, updated, classified, geocoded, or closed	Dispatch queue and detail, admin incidents, citizen view
/topic/missions	A mission is offered, accepted, progressed, cancelled, or completed	Dispatch, responder console
/topic/units	A unit's status or position changes	Dispatch map and unit lists, admin units
/topic/incidents/{id}/chat	A chat message is sent	Everyone viewing that incident's chat
/topic/hospital-transfers/{id}/vitals	Vitals are submitted for a transfer	Hospital dashboard

Deployment

The web console is deployed as a Next.js application and the API as a containerized Spring Boot service. For local use and reproduction, a single Docker Compose file starts PostgreSQL, the backend, and the web console together; configuration is supplied entirely through environment variables. The schema is created and evolved by Flyway migrations, and Hibernate is set to validate the schema at start-up.

7.2 Data Model

Conceptual entity-relationship model. Rectangles are entities, diamonds are relationships, and ovals are attributes. The figure is embedded at full resolution and can be enlarged in the digital copy; the class diagram (Section 7.3) and the data dictionary below give the same model in readable text form.

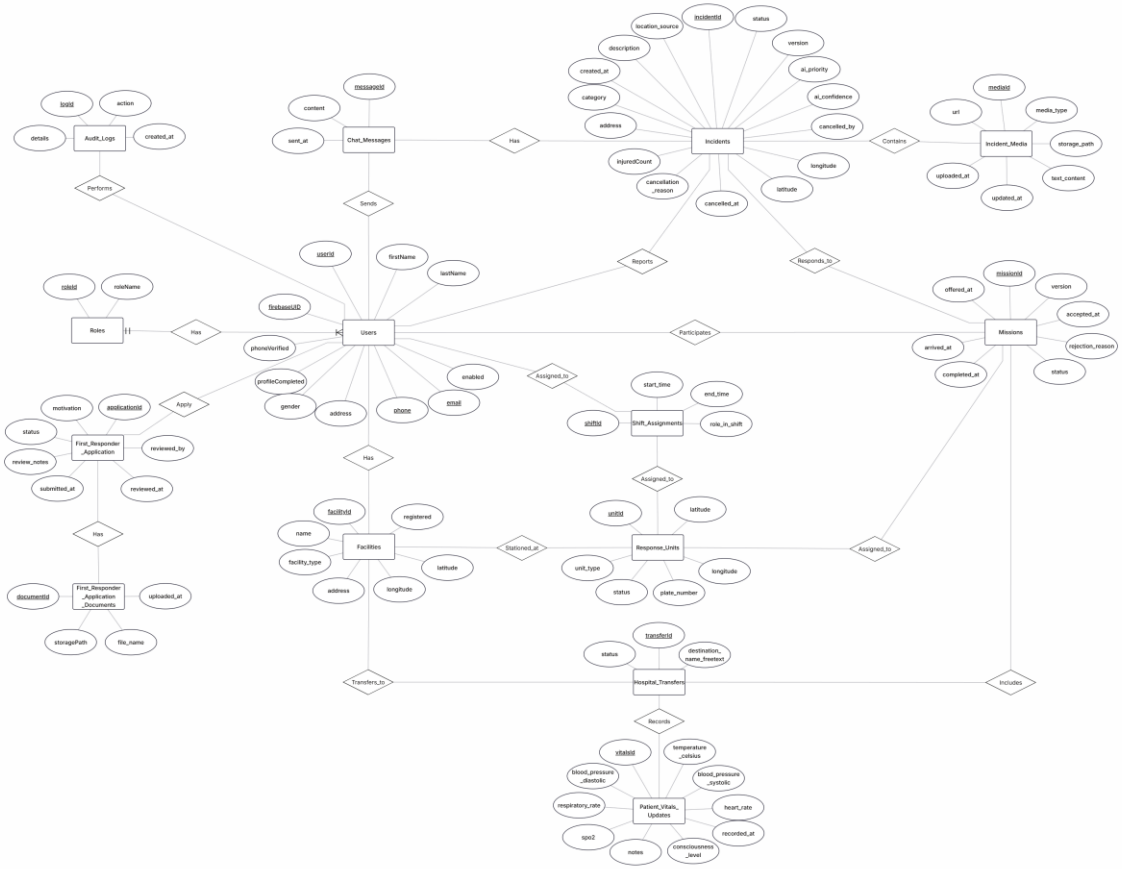


Figure 7.2 — Conceptual entity-relationship diagram

Physical schema as deployed in PostgreSQL, generated from the live database. Embedded at full resolution; the data dictionary that follows lists every column. Every foreign key shown is enforced; `reviewed_by` on applications and the actor and target columns on audit logs are deliberate soft references.

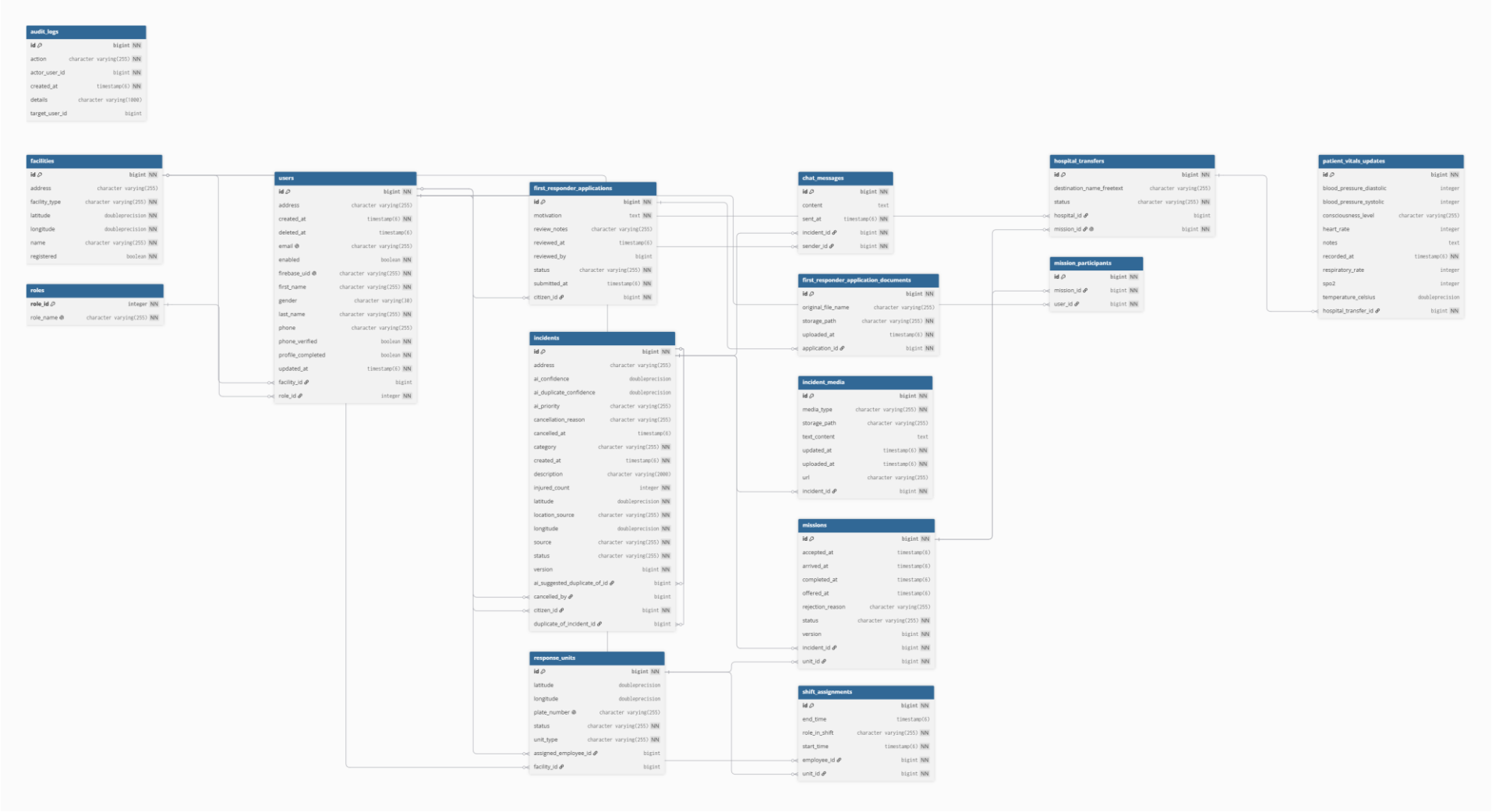


Figure 7.3 — Physical database schema

The schema enforces these notable constraints: user email and Firebase UID are unique; a unit's plate number is unique; each mission has at most one hospital transfer; every enumerated column is guarded by a check constraint. Phone-number uniqueness is deliberately not a database constraint — it applies only to verified numbers and is enforced in the application (Section 8, BR-18).

Data dictionary

Generated from the baseline migration. Enumerated columns list their permitted values. Columns that hold encrypted text are marked.

`roles`

Role catalogue (SUPER_ADMIN, ADMIN, DISPATCHER, HOSPITAL_STAFF, AMBULANCE_CREW, POLICE, FIREFIGHTER, FIRST_RESPONDER, CITIZEN).

Column	Type	Null	Constraints / notes
role_id	integer	no	PK
role_name	varchar(255)	no	UNIQUE

`users`

Every account. The Firebase UID links the row to the identity record; role and optional facility define access.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
address	varchar(255)	yes	
created_at	timestamp(6)	no	
deleted_at	timestamp(6)	yes	
email	varchar(255)	yes	UNIQUE
enabled	boolean	no	
firebase_uid	varchar(255)	no	UNIQUE
first_name	varchar(255)	no	
gender	varchar(30)	yes	one of: MALE, FEMALE
last_name	varchar(255)	no	
phone	varchar(255)	yes	
phone_verified	boolean	no	
profile_completed	boolean	no	
updated_at	timestamp(6)	no	
facility_id	bigint	yes	FK → facilities
role_id	integer	no	FK → roles

`facilities`

Unified hospitals, ambulance stations, fire stations, and police stations.

Column	Type	Null	Constraints / notes
id	bigint	no	PK

Column	Type	Null	Constraints / notes
address	varchar(255)	yes	
facility_type	varchar(255)	no	one of: HOSPITAL, AMBULANCE_STATION, FIRE_STATION, POLICE_STATION
latitude	double precision	no	
longitude	double precision	no	
name	varchar(255)	no	
registered	boolean	no	

response_units

Vehicles (and personal First Responder units) that can be assigned to incidents.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
latitude	double precision	yes	
longitude	double precision	yes	
plate_number	varchar(255)	yes	UNIQUE
status	varchar(255)	no	one of: AVAILABLE, BUSY, OFFLINE, MAINTENANCE
unit_type	varchar(255)	no	one of: AMBULANCE, FIRE_TRUCK, POLICE_CAR, FIRST_RESPONDER
assigned_employee_id	bigint	yes	FK → users
facility_id	bigint	yes	FK → facilities

shift_assignments

Who is on duty for which unit, and as LEAD or CREW.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
end_time	timestamp(6)	yes	
role_in_shift	varchar(255)	no	one of: LEAD, CREW
start_time	timestamp(6)	no	
employee_id	bigint	no	FK → users
unit_id	bigint	no	FK → response_units

incidents

The central record of an emergency, with AI output, location, lifecycle status, and duplicate links.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
address	varchar(255)	yes	
ai_confidence	double precision	yes	

Column	Type	Null	Constraints / notes
ai_duplicate_confidence	double precision	yes	
ai_priority	varchar(255)	yes	one of: CRITICAL, HIGH, MEDIUM, LOW
cancellation_reason	varchar(255)	yes	
cancelled_at	timestamp(6)	yes	
category	varchar(255)	no	one of: MEDICAL, POLICE, FIRE
created_at	timestamp(6)	no	
description	varchar(2000)	yes	
injured_count	integer	no	
latitude	double precision	no	
location_source	varchar(255)	no	one of: GPS, MANUAL_PIN
longitude	double precision	no	
source	varchar(255)	no	one of: CITIZEN_APP, SENSOR_DEVICE
status	varchar(255)	no	one of: NEW, AI_PROCESSED, DISPATCHER_REVIEW, ASSIGNED, IN_PROGRESS, RESOLVED, CANCELLED
version	bigint	no	
ai_suggested_duplicate_of_id	bigint	yes	FK → incidents
cancelled_by	bigint	yes	FK → users
citizen_id	bigint	no	FK → users
duplicate_of_incident_id	bigint	yes	FK → incidents

incident_media

Text and evidence attached to an incident; media is stored by private bucket path, never by public URL.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
media_type	varchar(255)	no	one of: PHOTO, VIDEO, AUDIO, TEXT
storage_path	varchar(255)	yes	
text_content	text	yes	encrypted at rest
updated_at	timestamp(6)	no	
uploaded_at	timestamp(6)	no	
url	varchar(255)	yes	
incident_id	bigint	no	FK → incidents

missions

One unit's assignment to one incident, with its own lifecycle.

Column	Type	Null	Constraints / notes
id	bigint	no	PK

Column	Type	Null	Constraints / notes
accepted_at	timestamp(6)	yes	
arrived_at	timestamp(6)	yes	
completed_at	timestamp(6)	yes	
offered_at	timestamp(6)	yes	
rejection_reason	varchar(255)	yes	
status	varchar(255)	no	one of: OFFERED, ACCEPTED, REJECTED, EN_ROUTE, ARRIVED, COMPLETED, CANCELLED
version	bigint	no	
incident_id	bigint	no	FK → incidents
unit_id	bigint	no	FK → response_units

mission_participants

Snapshot of the crew on shift when a mission was offered.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
mission_id	bigint	no	FK → missions
user_id	bigint	no	FK → users

hospital_transfers

The destination and progress of an ambulance mission's patient handoff (one per mission).

Column	Type	Null	Constraints / notes
id	bigint	no	PK
destination_name_freetext	varchar(255)	yes	
status	varchar(255)	no	one of: SELECTED, EN_ROUTE, ARRIVED
hospital_id	bigint	yes	FK → facilities
mission_id	bigint	no	UNIQUE; FK → missions

patient_vitals_updates

Time-stamped vitals transmitted during a transfer.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
blood_pressure_diastolic	integer	yes	
blood_pressure_systolic	integer	yes	
consciousness_level	varchar(255)	yes	one of: ALERT, VERBAL, PAIN, UNRESPONSIVE
heart_rate	integer	yes	
notes	text	yes	encrypted at rest

Column	Type	Null	Constraints / notes
recorded_at	timestamp(6)	no	
respiratory_rate	integer	yes	
spo2	integer	yes	
temperature_celsius	double precision	yes	
hospital_transfer_id	bigint	no	FK → hospital_transfers

chat_messages

Per-incident conversation; content is encrypted at rest.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
content	text	yes	encrypted at rest
sent_at	timestamp(6)	no	
incident_id	bigint	no	FK → incidents
sender_id	bigint	no	FK → users

first_responder_applications

A citizen's application to become a First Responder.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
motivation	text	no	
review_notes	varchar(255)	yes	
reviewed_at	timestamp(6)	yes	
reviewed_by	bigint	yes	
status	varchar(255)	no	one of: PENDING, APPROVED, REJECTED
submitted_at	timestamp(6)	no	
citizen_id	bigint	no	FK → users

first_responder_application_documents

Supporting documents attached to an application, stored by private bucket path.

Column	Type	Null	Constraints / notes
id	bigint	no	PK
original_file_name	varchar(255)	yes	
storage_path	varchar(255)	no	
uploaded_at	timestamp(6)	no	
application_id	bigint	no	FK → first_responder_applications

audit_logs

Record of privileged identity changes (actor, action, target, details).

Column	Type	Null	Constraints / notes
id	bigint	no	PK
action	varchar(255)	no	
actor_user_id	bigint	no	
created_at	timestamp(6)	no	
details	varchar(1000)	yes	
target_user_id	bigint	yes	

7.3 Class Design

The domain model below shows the persistent entities, their key attributes, and their associations. Diamonds mark composition (a child that cannot exist without its parent); dashed arrows mark optional or self references.

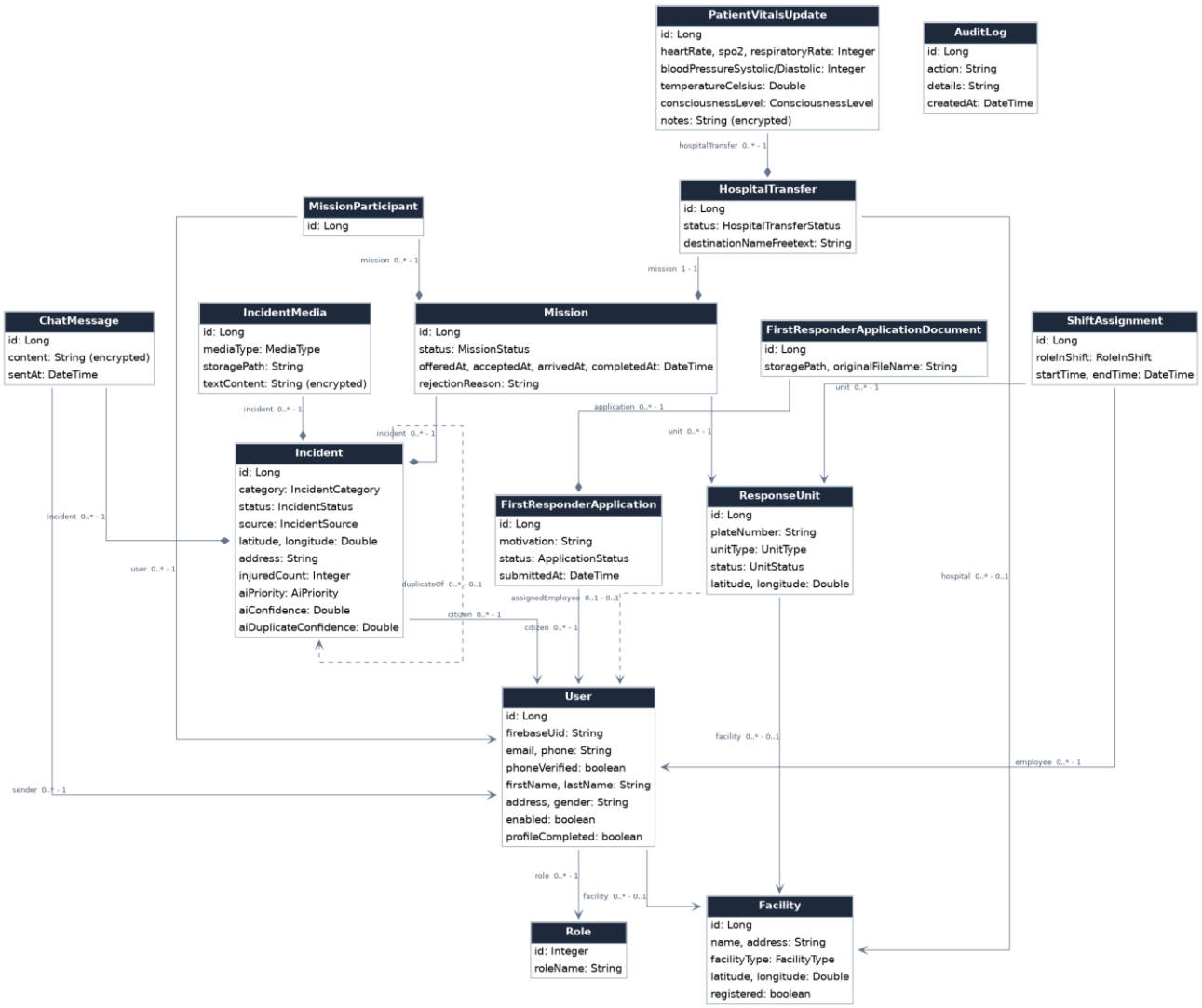


Figure 7.4 — Domain class diagram

Backend layering

The backend follows a strict controller → service → repository layering. Controllers translate HTTP to service calls and map exceptions to status codes; services hold all business rules and authorization; repositories are Spring Data JPA interfaces.

Area	Principal classes	Responsibility
Security	FirebaseTokenAuthFilter, SecurityConfig, WebSocketConfig	Verify tokens, resolve the user, open public endpoints, configure CORS and the WebSocket handshake
Identity	AuthServiceImpl, UserServiceImpl, ProfileCompletionEvaluator, FirebaseClaimsService, FirebaseSyncServiceImpl	Registration, profile rules, role claims, verified-only phone uniqueness, Firebase↔database reconciliation
Applications	FirstResponderApplicationServiceImpl	Atomic submission with documents, review, approval and personal unit creation
Incidents	IncidentServiceImpl, IncidentMediaServiceImpl, ChatServiceImpl	Submission, edit window, completion, duplicates, evidence, chat, access checks
Dispatch & field	MissionServiceImpl, ShiftServiceImpl, ResponseUnitServiceImpl	Assignment with row locking, mission lifecycle, shifts, unit status reconciliation, live location
Hospital	HospitalTransferServiceImpl, HospitalDashboardServiceImpl	Recommendation, selection, transfer progress, vitals, hospital view
Facilities	FacilityServiceImpl, FacilityDataSeeder, OverpassClient, RoleFacilityPolicy	Facility CRUD, OpenStreetMap seeding, role-to-facility-type policy
AI & geo	AiPriorityService, AiDuplicateDetectionService, ReverseGeocodingService	Asynchronous classification, duplicate suggestion, address resolution
Storage & crypto	SupabaseStorageService, ChatMessageContentConverter	Signed upload/download URLs; transparent AES-256-GCM field encryption
Real time	Incident-, Mission-, Chat-, Vitals-, and UnitEventPublisher	Publish state changes to STOMP topics

7.4 State Model

Three entities carry an explicit lifecycle. Each diagram shows the permitted transitions and the event that causes them.

Incident

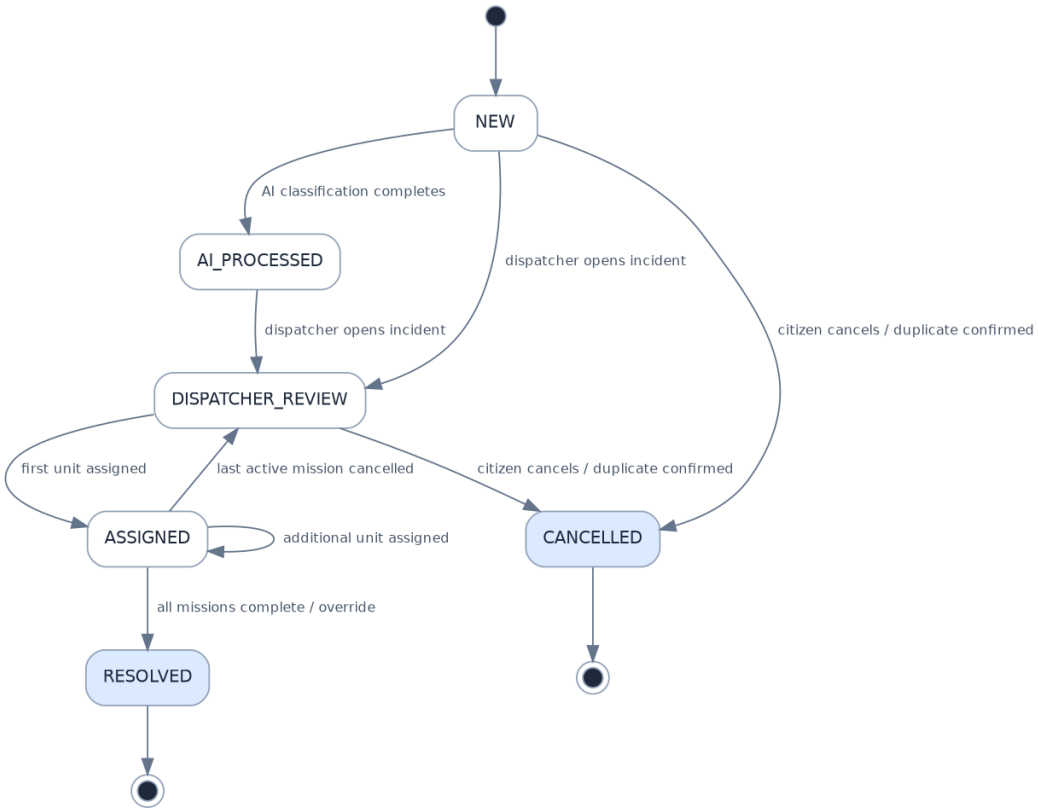


Figure 7.5 — Incident lifecycle

- A new incident is `NEW`; the AI result moves it to `AI_PROCESSED`, and a dispatcher opening it moves it to `DISPATCHER_REVIEW` (the AI step is not a prerequisite).
- The first unit assigned moves it to `ASSIGNED`; further units leave it there. Cancelling the last active mission returns it to `DISPATCHER_REVIEW`.
- `RESOLVED` is reached when every mission has completed, or by dispatcher override. `CANCELLED` is reached through a confirmed duplicate.
- `IN_PROGRESS` is part of the status enumeration but is reserved: no workflow currently enters it, because in-field progress is carried by the mission statuses.

Mission

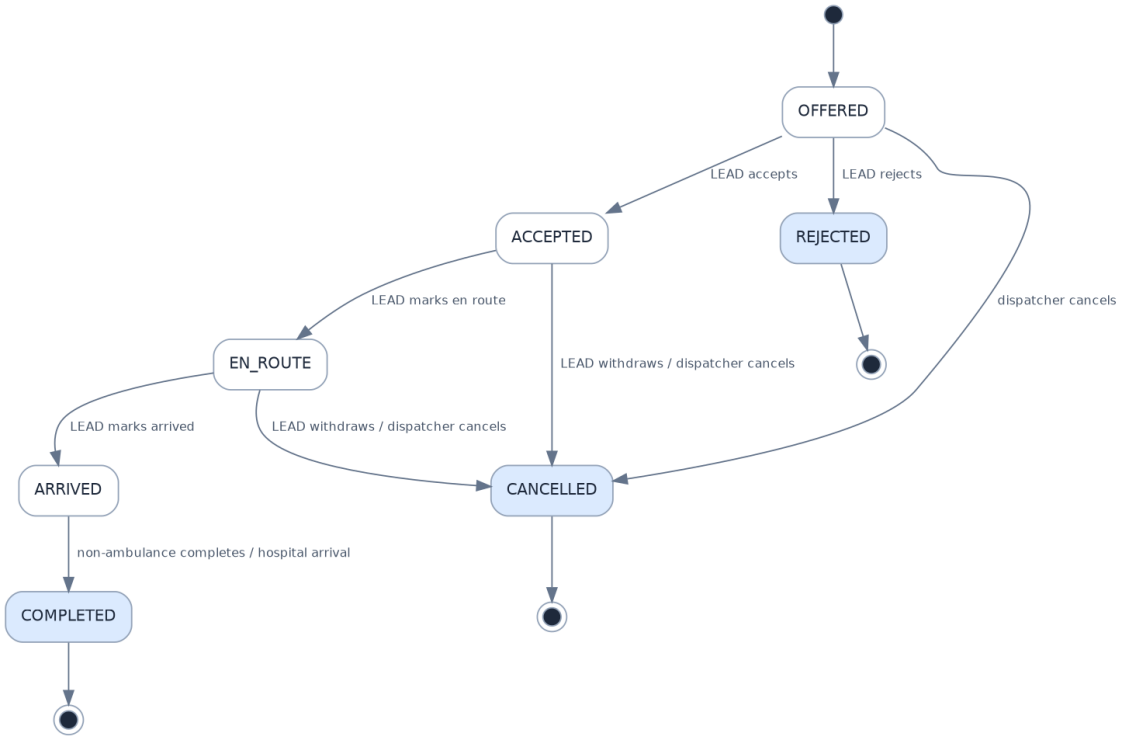


Figure 7.6 — Mission lifecycle

- Only the unit's LEAD accepts, rejects, progresses, withdraws, or completes. A dispatcher may cancel at any non-terminal stage.
- `COMPLETED`, `REJECTED`, and `CANCELLED` are terminal; each frees the unit.

Hospital transfer



Figure 7.7 — Hospital transfer lifecycle

Arrival at the hospital completes the parent mission and, when it was the last open one, resolves the incident.

7.5 Behavioral Views (Sequence Diagrams)

Incident submission, AI triage, and unit assignment

Submission returns as soon as the incident is stored. AI classification runs in the background and its result is pushed to dispatchers when ready. Assignment locks the unit row so that two dispatchers cannot offer the same unit.

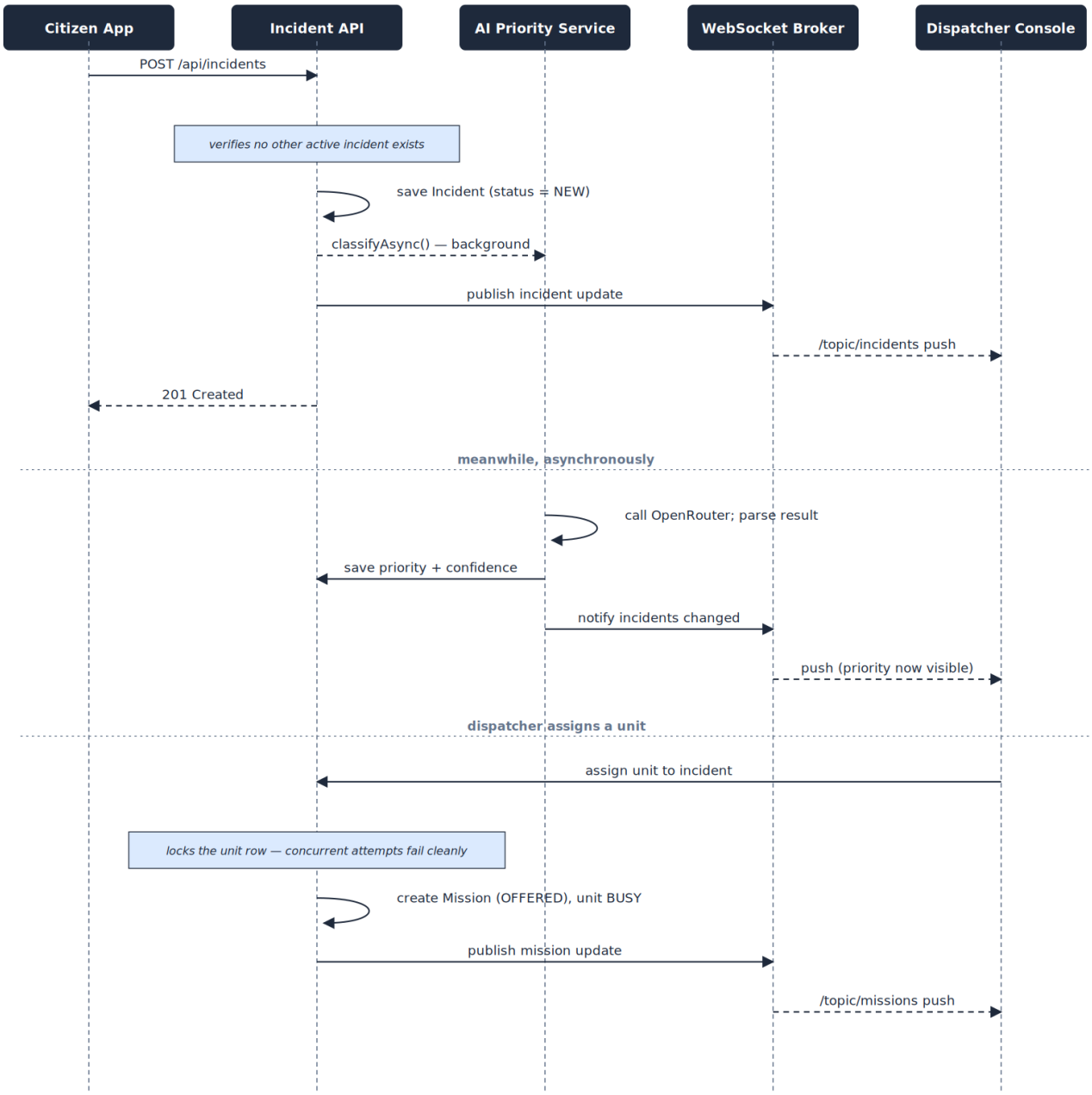


Figure 7.8 — Incident submission, AI triage, and assignment

Passwordless citizen registration

No account exists until the sign-in link is opened. The database record is created immediately after Firebase confirms the email; the profile gate then collects the remaining details and a password. Re-requesting a link for a half-registered account resumes at the profile step.

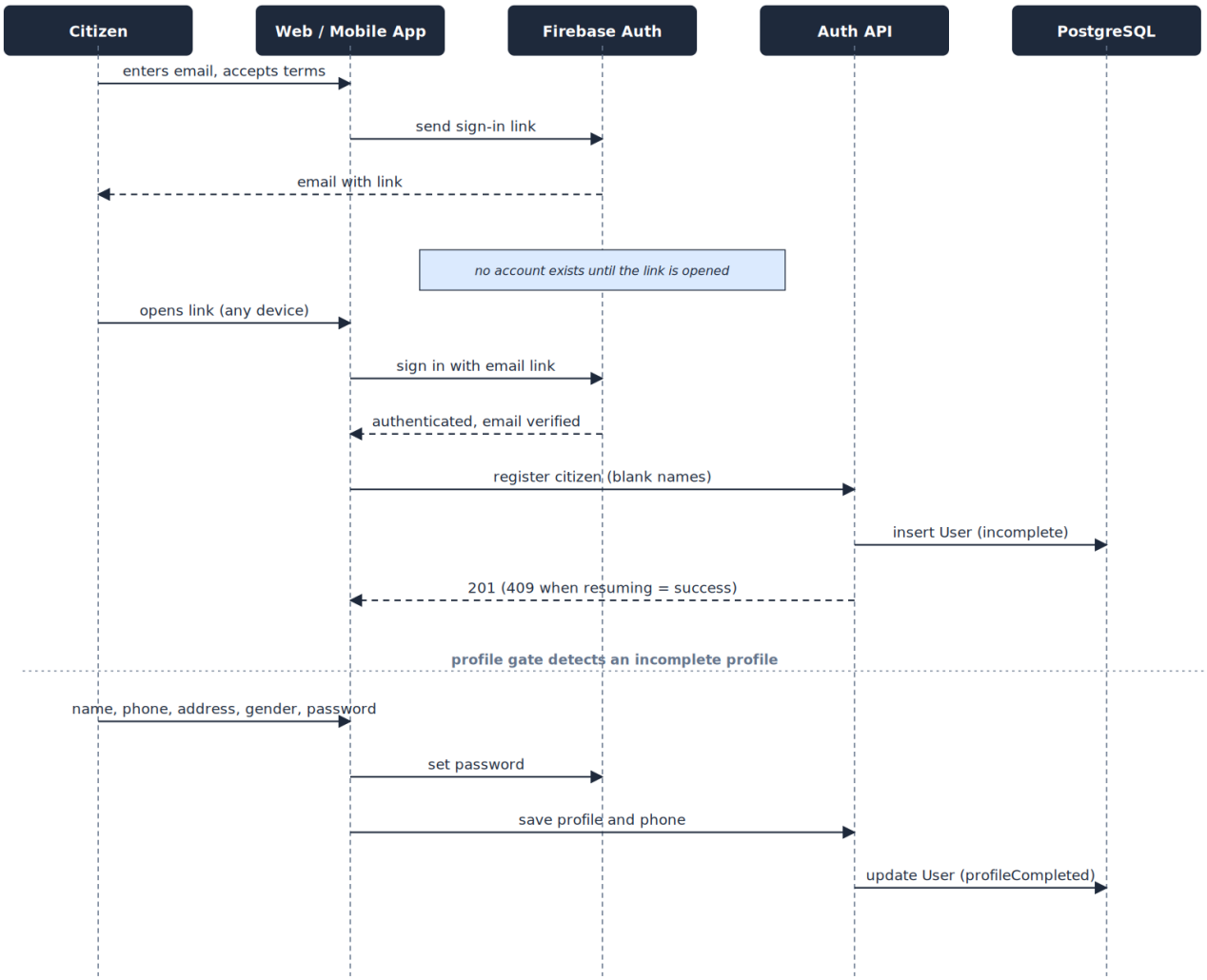


Figure 7.9 — Passwordless registration and profile completion

7.6 Authentication and Authorization Model

Token flow

- A user signs in with Firebase (email link, password, or Google) and receives an ID token.
- The web console additionally exchanges the token for an httpOnly session cookie used to gate page routes; the mobile app has no cookie and holds the token in memory.
- Every API call carries `Authorization: Bearer <ID token>`. A servlet filter verifies it with the Firebase Admin SDK, loads the NAJDA user (with role and facility), and exposes the role to Spring Security.
- Registration bootstrap, the email-availability check, the health endpoint, and the WebSocket handshake are the only endpoints reachable without a user record.

Authorization

Role checks are declared on controller methods and re-verified inside services, which also perform ownership checks (for example, that the caller is the unit's active LEAD, or took part in the incident being read).

Role	Console	Principal capabilities
Citizen	Emergency	Report and edit own incidents, chat, view own history, apply to become a First Responder
Dispatcher	Dispatch	Queue, assignment, cancellation, duplicates, completion override, retry AI, chat
Ambulance Crew · Police · Firefighter · First Responder	Responder	Shifts, missions, live location; ambulance additionally hospital handoff and vitals
Hospital Staff	Hospital	Incoming transfers and live vitals for the linked hospital
Admin	Admin	All dispatcher capabilities plus account, facility, unit, incident, and application administration
Super Admin	Admin	All admin capabilities plus Super Admin management, phone override, incident deletion, and Firebase reconciliation

External services

Service	Used for	If unavailable
Firebase Authentication	Sign-in, token verification, account administration, sign-in emails	Sign-in and registration are unavailable; existing sessions continue until token expiry
Supabase Storage	Private evidence and document storage via signed URLs	Uploads and viewing fail; incidents themselves are unaffected
OpenRouter (language model)	Priority classification and duplicate suggestion	Incident is stored without a priority; a dispatcher can retry
OSM Nominatim	Reverse geocoding	Address stays empty; coordinates remain authoritative; batch jobs stop on rate limiting
OSM Overpass	Facility seeding	Seeding fails over across mirrors, then reports an error
OSRM	Driving routes	Maps fall back to a straight line
EmailJS	Support form delivery	The form reports a failure and the user may retry

8 Business Rules

These rules govern system behaviour and are enforced on the server.

ID	Rule	Applies to
BR-1	A citizen may have only one incident in an active status (NEW, AI_PROCESSED, DISPATCHER_REVIEW, ASSIGNED, IN_PROGRESS).	UC-6
BR-2	An incident's coordinates must lie inside Egypt's bounding box.	UC-6
BR-3	A citizen may edit an incident, or add or remove its evidence, only while it is NEW, AI_PROCESSED, or DISPATCHER_REVIEW.	UC-17
BR-4	Evidence is uploaded to private storage through a signed URL and is registered only after the system confirms the file exists at that path; only permitted media types are accepted.	UC-6
BR-5	Signed download URLs expire after five minutes and are issued only after an authorization check.	UC-5, UC-6, UC-18
BR-6	AI output is advisory. A duplicate is suggested only for open incidents of the same category within 2 km and 120 minutes, at confidence 0.5 or higher. AI never assigns, cancels, or closes.	UC-13, UC-14
BR-7	Only AVAILABLE units can be assigned, and a unit is available only while a LEAD is on shift for it.	UC-7, UC-10
BR-8	Assignment locks the unit row; a unit is marked BUSY when a mission is offered and freed when the mission is rejected, cancelled, or completed.	UC-7, UC-8, UC-29
BR-9	Mission actions and hospital-transfer actions may be performed only by the unit's active LEAD (dispatcher cancellation excepted).	UC-8, UC-9
BR-10	When a mission is offered, the crew currently on shift is recorded as its participants.	UC-7
BR-11	A LEAD may withdraw only while a mission is ACCEPTED or EN_ROUTE; a dispatcher may cancel at any non-terminal stage. Cancelling the last active mission reopens the incident.	UC-8, UC-29
BR-12	A non-ambulance mission completes from ARRIVED; an ambulance mission completes when its hospital transfer arrives. An incident resolves when all its missions are complete; a dispatcher override completes them all.	UC-8, UC-9, UC-15
BR-13	An employee may hold one active shift. Facility-linked roles may use only units at their own facility; Ambulance Crew and First Responder are exempt, and a First Responder works only their own unit.	UC-10
BR-14	Hospital Staff must be linked to a hospital; Ambulance Crew may link to an ambulance station or hospital; Police to a police station; Firefighter to a fire station; other roles to none.	UC-2, UC-4
BR-15	A facility's type is fixed after creation; its registered flag is set automatically when staff or a unit is linked.	UC-21
BR-16	Recommended hospitals are those within 30 km of the incident, ranked by distance from the incident; the nearest is flagged. A free-text destination is allowed as a last resort.	UC-9
BR-17	Vitals are accepted only while the transfer is en route and only for a registered hospital.	UC-9
BR-18	A phone number is unique among verified numbers only. Unverified numbers may coexist; a verified number can never be claimed by another account.	UC-1, UC-30, UC-31
BR-19	Citizen phone numbers are Egyptian mobile numbers (+20 followed by ten digits starting 10, 11, 12, or 15).	UC-1, UC-31
BR-20	Passwords have at least eight characters with an uppercase letter, a lowercase letter, and a number.	UC-1, UC-31

ID	Rule	Applies to
BR-21	Only a Super Admin may create or modify another Super Admin; the last enabled Super Admin cannot be disabled; phone overrides are Super Admin only and audit-logged.	UC-2, UC-3, UC-30
BR-22	An application needs a verified email and phone, at least one document, and no other pending application; approval changes the role and creates the personal unit.	UC-5
BR-23	Only the citizen, dispatchers, admins, and responders on an active mission may send chat messages; anyone who ever took part may read the history.	UC-18, UC-19
BR-24	Chat content, incident free text, and vitals notes are encrypted at rest.	UC-6, UC-9, UC-19
BR-25	Marking a duplicate cancels the incident and records the cancellation reason, actor, and time.	UC-14
BR-26	A unit in use cannot be deleted. Reconciliation only returns BUSY units that have no active mission to AVAILABLE and never alters OFFLINE units.	UC-24, UC-25
BR-27	Facility seeding skips unnamed records and duplicates by coordinates, and a rate-limit response from the geocoder stops the run.	UC-22, UC-23
BR-28	Email changes take effect only after the confirmation link is opened.	UC-31

9 Testing Strategy

9.1 Approach

NAJDA is verified through scenario-based, end-to-end walkthroughs that exercise each use case with the role that performs it, supported by static guarantees built into the stack.

- **Scenario verification per role.** Each use case is walked through as the actor, including its alternate and exception flows, against a running stack.
- **Schema validation at start-up.** Hibernate validates entities against the Flyway-managed schema, so mapping drift fails the application on boot.
- **Type safety.** The web and mobile clients are strictly typed against the API contract, and both clients share the same contract types.
- **Authorization checks.** Each protected operation is exercised with a role that should be refused as well as one that should succeed.

9.2 Test Environment and Data

- A start-up initializer provisions one account per role, each backed by a test facility, so every console can be exercised immediately.
- Facilities are seeded from OpenStreetMap through the administration tools.
- Phone verification uses Firebase test phone numbers with fixed codes, so SMS delivery is not required.
- The full stack runs locally through Docker Compose, so every scenario can be reproduced from a clean checkout.

9.3 Verification Scenarios

ID	Scenario	Expected result	Use case
TC-1	Register with an email link, open it on the same browser	Account created with verified email; profile completion shown	UC-1
TC-2	Open the sign-in link on a different device	Email is requested once, then registration continues	UC-1
TC-3	Abandon registration, request a new link	Registration resumes at profile completion without error	UC-1
TC-4	Admin creates an employee with a facility-linked role	Account created, facility registered, reset link returned	UC-2
TC-5	Plain Admin tries to create or edit a Super Admin	Refused with access denied	UC-2, UC-3
TC-6	Citizen with unverified phone submits an application	Refused with a verification message	UC-5
TC-7	Approve an application	Role becomes First Responder; personal unit exists	UC-5
TC-8	Submit an incident with GPS and evidence	Incident NEW; dispatcher sees it live; address and AI priority appear later	UC-6
TC-9	Submit a second incident while one is active	Refused with a conflict	UC-6
TC-10	Edit description after a unit is assigned	Refused; before assignment it succeeds	UC-17
TC-11	Open a closed incident as a past responder	History, evidence, and chat are readable	UC-18
TC-12	Send a chat message as a non-participant	Refused	UC-19
TC-13	Two dispatchers assign the same unit	One succeeds; the other receives a conflict	UC-7

ID	Scenario	Expected result	Use case
TC-14	Cancel an accepted mission	Unit AVAILABLE; incident back to review if no other mission	UC-29
TC-15	Retry AI after a failed classification	Priority populates and is pushed live	UC-13
TC-16	Confirm and dismiss an AI duplicate	Confirm cancels and links; dismiss clears the suggestion	UC-14
TC-17	Force-complete an incident with two active units	Incident RESOLVED; both units AVAILABLE	UC-15
TC-18	Super Admin deletes an incident with a busy unit	Incident removed; unit freed	UC-16
TC-19	Start, join, leave, and end a shift	Statuses follow the rules; ineligible attempts are refused	UC-10
TC-20	A CREW member tries to accept a mission	Refused; only the LEAD may act	UC-8
TC-21	LEAD withdraws while en route	Mission CANCELLED; unit freed	UC-8
TC-22	Share location with permission denied	Status shows denied; dispatch uses the facility location	UC-12
TC-23	Select a hospital on the map and confirm	Transfer SELECTED; recommended and own-base indicators visible	UC-9
TC-24	Submit vitals for an unregistered hospital	Refused with an explanation	UC-9
TC-25	Hospital staff opens an incoming transfer	Map and live vitals update as readings arrive	UC-20
TC-26	Seed hospitals twice	Second run creates no duplicates	UC-22
TC-27	Geocoder rate-limits during seeding	Run stops cleanly; re-run later completes	UC-22, UC-23
TC-28	Create an ambulance unit	Only ambulance stations and hospitals are offered as home station	UC-24
TC-29	Run Firebase reconciliation	Orphans listed; sync creates an incomplete profile; deletes are guarded	UC-26
TC-30	Switch to Arabic	Layout flips to right-to-left; numbers and links stay left-to-right	UC-28

10 Requirements Traceability Matrix

Every functional requirement traces forward to a user story, a use case, and the component that implements it.

Req.	User story	Use case	Implementing component
FR-1	US-1	UC-1	AuthController.registerCitizen · AuthServiceImpl.registerCitizen
FR-2	US-2	UC-1	UserServiceImpl.updateProfile / setUnverifiedPhone · ProfileCompletionEvaluator
FR-3	US-1	UC-1	AuthServiceImpl.registerCitizen (existing UID → conflict handled by client)
FR-4	US-3	UC-1	AuthController email-availability endpoint · useEmailAvailability
FR-5	—	All	FirebaseTokenAuthFilter · SecurityConfig
FR-6	US-4	UC-2	AuthController.registerEmployee · AuthServiceImpl.registerEmployee
FR-7	US-4	UC-2, UC-4	RoleFacilityPolicy · FacilityServiceImpl.requireFacilityOfType
FR-8	US-5	UC-30	UserController · UserServiceImpl.getAllUsers
FR-9	US-5	UC-3	UserServiceImpl.disableUser / enableUser
FR-10	US-5	UC-4	UserServiceImpl · FirebaseClaimsService.syncRoleClaim
FR-11	US-6	UC-30	UserServiceImpl.adminOverrideEmail / adminOverridePhone · AuditLog
FR-12	US-5	UC-30	UserServiceImpl.resendPasswordReset
FR-13	US-6	UC-1, UC-30, UC-31	UserServiceImpl.setUnverifiedPhone / syncVerifiedPhone / adminOverridePhone
FR-14	US-7	UC-31	UserServiceImpl · Firebase client SDK (email, phone, password)
FR-15	—	All	@PreAuthorize on controllers · service-layer ownership checks
FR-16	US-8	UC-5	FirstResponderApplicationServiceImpl.submit
FR-17	US-9	UC-5	FirstResponderApplicationServiceImpl.approve / reject / deleteDocument
FR-18	US-6	UC-26	FirebaseSyncServiceImpl · FirebaseSyncController
FR-19	US-10	UC-6	IncidentController · IncidentServiceImpl.submit
FR-20	US-10	UC-6	IncidentServiceImpl.submit (active-incident check)
FR-21	US-11	UC-6	IncidentMediaServiceImpl · SupabaseStorageService
FR-22	US-12	UC-17	IncidentServiceImpl.updateInjuredCount · IncidentMediaServiceImpl
FR-23	US-13	UC-18	IncidentServiceImpl (mine, getById, requireCanView)
FR-24	US-15	UC-18	IncidentMediaServiceImpl.getDownloadUrl
FR-25	—	UC-6	ReverseGeocodingService
FR-26	US-14	UC-19	ChatController · ChatServiceImpl
FR-27	US-15	UC-19	ChatMessageContentConverter (chat, incident text, vitals notes)
FR-28	US-16	UC-7	IncidentServiceImpl queue / active · IncidentEventPublisher
FR-29	US-17	UC-7	MissionServiceImpl.assignUnitMission · web unit-assignment selection
FR-30	US-18	UC-7	MissionServiceImpl (pessimistic unit lock)
FR-31	US-19	UC-29	MissionServiceImpl.cancelMission

Req.	User story	Use case	Implementing component
FR-32	US-19	UC-15	IncidentServiceImpl.completeIncident
FR-33	US-21	UC-16	IncidentServiceImpl.deleteIncident
FR-34	US-17	UC-7	Live operations map (web) · OSRM routing
FR-35	US-20	UC-14	IncidentServiceImpl.markDuplicate / dismissDuplicateSuggestion
FR-36	US-32	UC-13	IncidentServiceImpl.retryAiPriority
FR-37	US-22	UC-10	ShiftServiceImpl · ShiftController
FR-38	US-22	UC-10	ResponseUnitServiceImpl.getUnitsForFacility
FR-39	US-23	UC-8	MissionServiceImpl.acceptMission / rejectMission
FR-40	US-24	UC-8	MissionServiceImpl (en route, arrived, complete, withdraw)
FR-41	US-25	UC-12	ResponseUnitServiceImpl.updateLocation · location-sharing hook
FR-42	US-26	UC-11, UC-25	ShiftServiceImpl (force end) · ResponseUnitServiceImpl.reconcileStatuses
FR-43	US-25	UC-8	Mission detail and past-missions views
FR-44	US-27	UC-9	HospitalTransferServiceImpl.getRecommendedHospitals
FR-45	US-27	UC-9	Hospital selection map and transfer panel
FR-46	US-30	UC-9	HospitalTransferServiceImpl (selection and progression)
FR-47	US-28	UC-9	HospitalTransferServiceImpl.sendVitalsUpdate · VitalsEventPublisher
FR-48	US-29	UC-20	HospitalDashboardServiceImpl · hospital feed
FR-49	US-16	UC-13	AiPriorityService
FR-50	US-31	UC-14	AiDuplicateDetectionService
FR-51	US-31	UC-13, UC-14	AI services write advisory fields only
FR-52	US-33	UC-21	FacilityServiceImpl · FacilityController
FR-53	US-33	UC-22	FacilityDataSeeder · OverpassClient
FR-54	US-33	UC-23	FacilityServiceImpl.backfillMissingAddresses
FR-55	US-34	UC-24	ResponseUnitServiceImpl · unit form
FR-56	US-35	UC-14, UC-15	Admin incidents feed and detail panel
FR-57	US-36	UC-28	next-intl · react-i18next · language switchers
FR-58	US-37	UC-27	Support page (EmailJS)
FR-59	US-38	UC-1, UC-6, UC-8, UC-10	Expo mobile application

11 Assumptions and Constraints

11.1 Assumptions

ID	Assumption
A-1	NAJDA operates in Egypt: phone numbers use +20, incident coordinates are validated to Egypt's bounding box, and facility data is drawn from Egyptian OpenStreetMap records.
A-2	Citizens can supply a location, by device GPS or by placing a pin on the map.
A-3	Users have connectivity sufficient for live updates; the interface remains usable without them but refreshes less promptly.
A-4	More than one environment (for example local development and a hosted instance) may share one Firebase project, which is why account reconciliation tooling exists.
A-5	The platform is a simulation: no report dispatches a real unit and nothing is connected to a real emergency service.

11.2 Constraints

ID	Constraint
C-1	Free public services (Nominatim, Overpass, the public OSRM server, OpenFreeMap, free language models) impose usage limits and variable availability. The platform degrades gracefully around each, and geocoding batches are paced to a one-request-per-second policy.
C-2	Live updates use an in-memory STOMP broker inside a single API instance; running several instances would require a shared broker.
C-3	A hosted free-tier API may be idle between uses, so the first request after a quiet period can be slow.
C-4	Email uniqueness within the password and email-link providers is enforced by Firebase and cannot be relaxed; the registration flow is designed around that.
C-5	On mobile, phone verification supports Firebase test numbers, and the mobile client covers the Citizen and Responder roles; use cases marked Web are available on the web console.
C-6	The mobile application is distributed as a signed Android package through GitHub Releases.

12 Design Decisions and Future Work

12.1 Key Design Decisions

Decision	Rationale
Firebase is identity only; Postgres owns authority	A managed provider handles credentials and verification, while roles, facility links, and history live in one database that every permission check can trust.
Create the account record only after email confirmation (passwordless link)	Firebase refuses a second password account for an address, so an abandoned unverified sign-up could lock out the real owner. Deferring creation until the link is opened removes the problem entirely.
WebSocket push over an in-memory broker	Meets the live-update requirement with no extra infrastructure for a single-instance deployment; a shared broker is a contained change if scale ever requires it.
One Facility entity with a type	Hospitals and stations share location, address, and registration semantics; one table removes parallel code paths.
A shift is the only evidence a unit has an operator	Assignment is gated on a LEAD being on duty, so dispatch never offers work to an unattended vehicle.
Phone uniqueness only among verified numbers, in the application	A blanket constraint lets an unverified entry block the real owner. Enforcing at verification time protects owners while allowing unverified overlap.
Private storage with short-lived signed URLs	Media is sensitive; no link outlives its purpose, and every view is an authorization decision.
Server-held field encryption, not end-to-end	Dispatchers and administrators must be able to read conversations for oversight, so protection targets database exposure rather than the application itself.
Hospital recommendation measured from the incident	The incident location always exists and does not move, unlike a unit's possibly stale GPS fix.
Locale in a cookie, not the URL	NAJDA is an authenticated tool, so per-language URLs for search indexing add cost without benefit.
Versioned migrations with schema validation	Two environments share one schema; numbered migrations apply identically to both and validation fails loudly on drift.
AI is advisory	Priority and duplicate output inform a human decision and are always dismissible.

12.2 Future Work

Wearable and sensor integration

The data model is already prepared for incidents that originate from a device rather than a person: each incident carries a `source` (currently `CITIZEN_APP`, with `SENSOR_DEVICE` reserved). The planned extension lets a citizen pair a wearable, such as a smartwatch, with their account. When the device detects a health event — for example a fall or an irregular heart rhythm — it reports to the backend, which would either attach the reading to the citizen's current incident or open a medical incident of its own, then run it through the same triage and dispatch flow.

- **Pairing and authentication.** A citizen pairs a device from their account and receives a long-lived, revocable device key, since a wearable cannot perform an interactive sign-in.
- **Structured readings.** Device reports carry an event type, measurements such as heart rate, and a location, stored as structured records linked to the incident rather than flattened into text.
- **Active-incident handling.** A report from a citizen who already has an active incident updates that incident as an urgent change instead of creating a competing one.

Appendix A Technology Stack

Layer	Technology	Purpose
Backend	Java, Spring Boot	REST API, STOMP WebSocket, application services
	Spring Security, Spring Data JPA / Hibernate	Authentication filter, method-level authorization, persistence
	PostgreSQL, Flyway	System of record and versioned schema migrations
	Firebase Admin SDK	Token verification, account and claim management
	Docker, Docker Compose	Container image and one-command local stack
Web console	Next.js (App Router), React, TypeScript	Role-scoped console for every role
	Tailwind CSS v4	Styling, dark mode, logical (RTL-aware) utilities
	TanStack Query, Zustand	Server state and local state
	next-intl	English / Arabic with right-to-left layout
	MapLibre GL, @stomp/stompjs	Maps and live updates
	Firebase JS SDK, EmailJS	Sign-in; support form delivery
Mobile app	Expo SDK 57, Expo Router, React Native, TypeScript	Citizen and Responder companion app
	NativeWind v4, react-i18next	Styling; English / Arabic with RTL
	expo-location, react-native-webview + MapLibre GL JS	Location tracking; live maps
External services	Firebase Authentication	Identity
	Supabase Storage	Private media bucket with signed URLs
	OpenRouter	Language-model access for triage
	OpenStreetMap: Overpass, Nominatim; OSRM; OpenFreeMap	Facilities, geocoding, routing, map tiles
Hosting	Vercel (web), Render (API)	Hosted deployment

Appendix B API Reference

The API is REST over JSON under `/api`, with a STOMP endpoint at `/ws`. Errors return a JSON body `{ "error": "…" }` with status 400 (invalid input), 401 (unauthenticated), 403 (not permitted), 404 (not found), or 409 (conflict with current state). Authentication is `Authorization: Bearer <Firebase ID token>`.

Base path	Operations	Access
/api/auth	Register citizen (bootstrap after sign-in); register employee; current user; email availability	Bootstrap and availability public; employee creation Admin; current user any user
/api/users	Search and paginate; update profile; change role; enable / disable; link facility; override email; override phone; sync verified phone / email; generate reset link	Self for own profile; Admin for administration; Super Admin for phone override
/api/facilities	List (optionally by type); create; update; address backfill	Read: authenticated · Write: Admin
/api/admin/facilities	Seed facilities by type from OpenStreetMap	Admin
/api/units	List; units for my facility; create; update; delete; report location; reconcile statuses	Admin for management; LEAD for location
/api/shifts	Start; join; leave; end; current shift; force-end a unit's shift	Responder; force-end Admin
/api/incidents	Submit; mine; queue; active; all; by id; edit injured count; retry AI priority; mark duplicate; dismiss duplicate suggestion; complete; delete	Citizen for own; Dispatcher / Admin for operations; delete Super Admin
/api/incident-media	Create signed upload URL; register evidence; signed download URL; edit text; delete	Citizen for own incident; participants for viewing; Admin override on delete
/api/incidents/{id}/chat	List messages; send message	Participants (send restricted to active participants)
/api/missions	Assign unit; mine; by incident; accept; reject; en route; arrived; complete; withdraw; cancel	Dispatcher for assign / cancel; unit LEAD for the rest
/api/hospital-transfers	Recommended hospitals; select destination; mark en route / arrived; submit vitals; transfer by mission; transfers by hospital	Ambulance LEAD; hospital staff for their hospital
/api/hospital-dashboard	Incoming transfers for the caller's hospital	Hospital Staff
/api/first-responder-applications	Submit (multipart with documents); mine; all; approve; reject; document download URL; delete document	Citizen for own; Admin for review
/api/admin/firebase-sync	Report; create profile from Firebase account; delete orphaned Firebase account; delete orphaned database user	Super Admin
/ws	STOMP endpoint with topic subscriptions (Section 7.1)	Handshake public; topics carry non-sensitive change notices
/actuator/health	Health probe	Public

Appendix C Glossary

Term	Definition
Incident	The central record of an emergency, from report to closure.
Mission	One response unit's assignment to one incident, with its own lifecycle.
Response unit	A vehicle — or a First Responder's personal unit — that can be assigned to incidents.
Shift	The period during which an employee is on duty for a unit, as LEAD or CREW.
LEAD / CREW	The accountable operator of a unit's shift, and a crew member riding along.
Facility	A hospital, ambulance station, fire station, or police station.
Registered facility	A facility with real staff or units linked to it.
Hospital transfer	The destination and progress of an ambulance's patient handoff.
Vitals	Patient measurements (heart rate, blood pressure, SpO2, respiratory rate, temperature, consciousness) transmitted during a transfer.
Triage	Ranking an incident's urgency; here assisted by a language model and always reviewed by a person.
Passwordless link	A one-time sign-in link emailed to the user that confirms they control the address.
Profile gate	The application check that holds a user at profile completion until their account is complete.
Signed URL	A time-limited link, issued after an authorization check, that grants access to one stored file.
Orphaned account	An account that exists in Firebase or in the database but not in the other.
Migration	A numbered, versioned database change applied identically in every environment.